

Foil Design Tool for Onshape

User guide, design notes and parameter reference. Build 141 (wing 1.020, stabiliser 4.002, performance 1.013, mast 2.008, fuselage 4.012, mold 2.069).

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Two things before anything else. First, this is a modelling exercise, not the advice of a foil builder. The tool started as a side quest to the B-series propeller blade generator: sizing a prop needs a drag curve, a drag curve needs a foil, and a nine-hour flight turned that into six features. Second, every number the tool produces is a steady-state, constant-value estimate: one speed, one weight, flat water, a rider who doesn't move. Real foils live in chop, pumping, turns and landings, and they break on peaks and repetition. Trust the trend from one variant to the next; treat absolute numbers, stall speed especially, as optimistic until you have calibrated them against something you have ridden (see 10.1).

This document accompanies the *Foil design* Feature Studio. It explains what the tool does, how the six features work together, what every parameter means and how to use it, and the hydrofoil design and construction knowledge that was worked out while building it. It is written for a rider who designs and builds their own foils: pump, SUP and surf, wing, windfoil, kite and eFoil are all first-class targets.

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1. What the tool is

The tool is a single Onshape Feature Studio containing six custom features. Used in order in a Part Studio, they turn a handful of design decisions into a complete hydrofoil: front wing, stabiliser, mast with an optional motor pod, round tube fuselage, a full performance and stability analysis, and a set of 3D-printable molds with a layup schedule and fabric cutting templates.

Feature	What it makes	Reads	Publishes
Foil wing	The front wing, optionally with a tube socket and nacelle for a round fuselage	nothing	area, span, MAC, incidence, zero-lift angle, the analysis model, the socket position
Foil stabiliser	The rear wing, placed by Arm and Height behind the wing, optionally linked to the wing, with the production-style mount pad, tail fairing and bolt holes	the wing (link ratios, the tube socket axis the seat Height is measured from)	area, position, the mount contour and bolt pattern
Foil performance	No geometry. A speed sweep of lift, drag, trim, stability, structure, propulsion and energy	wing, stab, mast, fuselage	the spanwise bending loads used by the mold's stiffness check
Foil mast	The mast: tapered, thickened at the top, with a root fillet into the board plate, optional motor pod with a placeholder cavity, plate with bolt holes	the wing pivot (position only)	the mast law and plate for the mold and the fuselage pocket
Foil fuselage	The round tube: spigot into the wing socket, seat cut to the stab's mount contour, tapped holes matching the stab's bolts, mast pocket	wing socket, stab mount, mast	the fuselage shape for performance
Foil mold	Two-part (mast: three-part) printable molds for the wing, stab or mast: parting curtain, locator keys, pinch-off groove, undersized bolt cones, plugs, printer tiling, plus the layup, material and resin estimate, stiffness check and cut-template plate	the part and every model above	nothing

Frame and conventions. X points downstream (leading edge to trailing edge), Y along the span, Z up. The wing's root pivot, the quarter chord by default, sits at the world origin. Sections are streamwise. Incidence is positive leading edge up. *Decalage* is wing incidence minus stab incidence. Span is the projected tip-to-tip distance seen from above. Area is projected by default and always reported.

Design philosophy, as decided during development.

- Everything is parametric and mathematical: no sketches to draw, no freehand surfaces. Change a number, regenerate, read the notice.
- Foil area is a first-class input. You choose an area and an aspect ratio; the planform is solved to hit them exactly.
- The stab's arm and height are design decisions that belong with the wings. The mast and fuselage are structural parts that read those positions and never set them. The one deliberate exception is the stab's *Height*, which is measured from the wing's tube socket axis so its mount always lands on the fuselage.
- Discipline-neutral: nothing assumes an eFoil. Every tooltip gives typical values by discipline.
- Foils are carbon laid up in printed molds, not printed parts. Structural advice assumes carbon: woven skins, unidirectional caps and carbon tow.

- Every parameter has a short label so its value stays visible in the dialog, and a tooltip with limits, typical values and what the parameter does.
- Warnings tell you which field in which feature fixes the problem.

2. Quick start

1. In a Part Studio, open *Add custom features* and search for "Foil", or copy the public document and use its Part Studio. This guide is the PDF in that same document: right-click any of the features and choose *Open linked document* to find it again.
2. Add **Foil wing**. Pick a discipline-appropriate area, aspect ratio and section; leave the rest at defaults. The notice shows the area achieved, span, MAC and root chord.
3. Add **Foil stabiliser**. Set Arm, and Height as the seat offset from the fuselage centreline (−4 mm with the pad on top). Leave *Link to wing* on to size the stab as a fraction of the wing; the decalage is then set by *Link decalage*. Turn on the *Mount pad* if the stab will bolt to a tube fuselage.
4. Add **Foil performance**. Enter rider plus board mass, water, ride depth and the speeds you care about. Read the notice: L/D and power at cruise, stall, take-off margin, trim CG, static margin, root bending and cap thickness. Turn on *Log inputs* and open the FeatureScript console for the full sweep as CSV.
5. Add **Foil mast** and **Foil fuselage** when you want the structure. The performance feature picks up their drag automatically, so *Other cm²* is only for fittings.
6. Iterate. Change one or two things at a time and compare notices. The console lines (`WING_INPUTS`, `STAB_INPUTS`, and so on) capture every input and are what you paste when asking for help, whether on a forum or to the AI assistant in section 10.3.
7. Add **Foil mold** for each part you will build, select the part, set your printer's bed size, and read the layup and stiffness lines. Turn on *Cut templates* to get a plate whose top face exports as one DXF of every fabric shape.
8. Keep any custom geometry that the mold must follow, such as a hand-modelled motor cavity in the pod, *above* the mold feature in the feature list. The mold cuts its cavity from the part as it stands.

3. The six features

3.1 Foil wing

The wing is built from an area-first planform solver: you give target area and aspect ratio (or span and root chord), the chord distribution shape (superellipse of exponent n , or linear taper), and the solver finds the root chord that hits the area exactly. On top of that: an alignment line (which chord fraction stays straight in plan), power-law sweep, tip drop with an anhedral exponent and optional root dihedral, washout, a thickness distribution from root to tip, incidence about a chosen pivot, a tapered or rounded tip cap, and a minimum tip thickness.

Sections. NACA 4-digit is fully parametric (camber, camber position, thickness). Three published sections are built in as digitised coordinate tables: Eppler E817, Eppler E818 and NACA 63-412. A tabulated profile fixes its own camber; Root and Tip thick % scale its thickness. The trailing edge is held at a real thickness for moldability and thinned automatically near the tip so it never exceeds 3 % of the local chord.

Section polar. Analytic (thin-aerofoil theory) is the default. *Polar summary* takes thirteen numbers fitted from an Xfoil or XFLR5 run at two Reynolds numbers; *Built-in* uses stored polars for the tabulated sections.

With a polar, stall and profile drag come from the real section rather than from the Section CL max stand-in.

Tube socket. For a round fuselage the wing grows a nacelle with a socket for the tube: tube diameter and clearance, wall, blend and rear extension, depth, and cross pins. The socket's axis is what the fuselage spigots into and what the stab's Height is measured from.

Notice. Area achieved and definition, span, aspect ratio, MAC, root and tip chord, tip cap, thickness, incidence and washout, and the socket if on.

3.2 Foil stabiliser

The stab shares the wing's geometry core, so every planform, section, twist and tip control is the same. What is specific to the stab:

Position and link to wing. Arm is the distance along X from the wing pivot to the stab pivot, the pitching moment arm. Height is the vertical offset. *Link to wing* derives area, aspect ratio, thickness and incidence from the wing's reported values through an area ratio, an AR offset, a thickness offset and a decalage, so the stab follows when the wing changes.

Height is where the seat sits. When the wing has a tube socket, *Height* is the offset of the stab's seat plane, the pad face or the bare root surface, from the fuselage centreline, positive up; 0 puts it on the centreline. The stab is built and then moved so its seat lands exactly there, so resizing or relinking it never moves the joint. With the pad on top, 0 cuts the fuselage flat to half the tube and -4 mm is typical; with the pad underneath (the stab sits on top of the fuselage), $+4$. A seat plane below the tube is also valid: the fuselage's block carries the stab and the tube stays round. Without a socket on the wing, Height is the pivot's offset from the wing pivot. The notice and the `pivotZ_mm` log field report where the pivot ended up.

Mount pad: a fixed interface. Modelled on the Axis rear wing after studying a real one: a flat pad face on the fuselage side, faired up from the leading edge, with a tail fairing behind it that forms the lower half of the fuselage's tail. The mount is pinned to the stab pivot, not to the root section: *Flat start* is where the seat flat begins, *Tail start* is where the fairing and the fuselage tip begin to taper, *Tail end* is where both finish, *Bolt 1 at* and *Bolt pitch* place the bolts; all in millimetres from the pivot. Only the ramp from the leading edge to Flat start adapts to the stab. The fairing stays full width from the trailing edge to Tail start, so a small stab simply has a longer tail. Every stab with the same Arm therefore presents the same seat plane, flat length, tapped-hole pattern and fuselage tip: any stab fits any fuselage built for that arm. The notice warns and holds the geometry when a section does not fit the interface.

Mount bolts. Through holes on the centreline, chosen by *Bolt* size (M4 to M10, or custom hole sizes). The stab owns the pattern: it cuts the close-fit clearance hole and, with *Countersink* on, a 90° countersink to the ISO 10642 head diameter on the face away from the pad; the fuselage taps blind holes at the same positions with the matching tapping drill, read from the stab so the two can't disagree; the mold puts undersized cones there to be drilled and countersunk after cure.

Bolt	Clearance hole	Tapping drill	Countersunk head
M4	4.4	3.3	8.96
M5	5.4	4.2	11.2
M6	6.4	5.0	13.44
M8	8.4	6.8	17.92
M10	10.5	8.5	22.4

The mast plate and the wing socket's cross pin use the same dropdown, with the clearance 0.1 mm looser because several holes have to line up across two parts.

Notice. Planform summary, the mount description with the fixed-interface positions, the bolt holes, the seat report, and with a wing present: area as a percentage of the wing, tail volume, geometric and aerodynamic decalage.

3.3 Foil performance

Builds no geometry. It reads every model above and runs a speed sweep from take-off to max speed:

- **Lifting line** for the wing and the stab (span efficiency, lift slope, induced drag), the classic far-field downwash at the stab, and a trim solution at each speed: the body angle and the stab load that carry the mass in level flight.
- **Profile drag** from flat-plate skin friction with a thickness form factor on the real wetted area, plus a lift-dependent term, or the section polar's own drag bucket when one is given.
- **Mast, fuselage and pod drag** computed from the Foil mast and Foil fuselage features: submerged mast friction plus spray where it pierces the surface, pod friction on its wetted fraction, fuselage body friction. *Other cm²* is only for junctions, fittings and bolt heads.
- **Stall** where the first section inboard of 95 % span reaches the section's CL max; **take-off margin** as a percentage of CL max at the take-off speed.
- **Trim CG** (where the rider's mass must sit for level flight) and **static margin** (its distance ahead of the neutral point, as a percentage of MAC) at every speed.
- **Structure**: half-wing root bending at the load factor, the UD cap thickness needed for the chosen cap stress and width, and the spanwise moment distribution published for the mold's stiffness check.
- **Propulsion** (optional): Wageningen B-series open-water propeller, motor kV, cells, gearing, current limit and winding resistance, pack capacity. It solves thrust equals drag at every speed and reports rpm, throttle, current, battery watts, Wh/km, range and top speed, plus the powered trim: the thrust couple, the powered trim CG and the powered static margin.
- **Layout**: from Pod height it decides whether the foil is an eFoil (the pod can never clear the water at a rideable depth) or foil assist (the wing can ride with the pod out), and reports both modes.

Notice. One line: cruise L/D, drag and power, stall, take-off as a percentage of CL max, trim CG and static margin, root bending and caps; with propulsion, rpm, throttle, battery watts, Wh/km, range and top speed, and the powered trim. Warnings turn the notice yellow: stall at take-off, take-off within 15 % of stall, stab CL above 0.8, cavitation number below 1.5, depth under 2 MAC, propulsion out of limits, powered static margin under 5 %.

Console. The full report and a CSV with one row per speed. Its columns are listed in chapter 9.

3.4 Foil mast

The mast is a loft of symmetric NACA 00xx sections along a *mast law*: chord runs linearly from the base width to *Top chord*, with the taper on the leading edge, the trailing edge or both; thickness is constant to *Thicken from %* and then eases with a smoothstep to *Top thick* at the plate. A straight leading edge moves the plate aft relative to the wing, a straight trailing edge the reverse, so the taper side also moves the rider relative to the wing.

Root fillet. A fillet all round the section where it meets the plate's underside, held inside the plate with 2 mm to spare. Onshape will fillet the section's surface but not the 0.8 mm trailing edge line, so the fillet stops there and leaves a notch of trailing edge thickness; the console line `MAST_FILLET` reports the seam edges and the attempt that worked.

Motor pod. A streamlined pod on the mast for eFoil: diameter, height on the mast, where the nose starts, fairing length, blend height, aft extension, and *Pod tilt*, which rotates the pod about the span axis so the thrust vector can be aimed. The pod carries a placeholder cavity from its rear face (diameter and length parametric) that you are meant to replace or extend by hand with the real motor shape; the mold shapes its plug from whatever is there.

Plate. The board plate as a block with two rows of bolts on the track spacing, chosen by *Bolt* size (M8 by default), countersunk from below to the standard head diameter.

Notice. Height, chord and taper, thickness law, root fillet result, pod summary, plate.

3.5 Foil fuselage

A round tube with a spigot into the wing's socket, a body of the socket's tube diameter running aft, then a tail. With a stab mount present the tail is *seated*: the underside is cut to the stab's published mount contour (ramp then flat), the tip tapers from the stab's Tail start to its Tail end, blind tapped holes are cut at the stab's bolt positions to the chosen depth with the tapping drill for the stab's bolt, and a block under the flat fills the space when the seat is deeper than the tube. A pocket for the mast base and cross-pin holes matching the wing's socket pins are optional. The notice reports the seat, the holes made and any conflict, naming the field in the other feature that fixes it.

3.6 Foil mold

Select the finished part (wing, stab or mast). The mold feature:

- **Parts the mold** along a curtain built from the camber line's silhouette at up to 80 stations, with *Tip bias* packing stations toward the tip where the surface curls fastest. For the stab mount the parting steps onto the pad plane. The mast is molded flat in three parts: left and right halves plus a tray that forms the plate top, with straight locator pins. A stab with its pad underneath is molded upside down, as its mirror image, and turned over after cure; the notice says so.
- **Block and flange** of chosen size, **locator keys** (printed frustums or drilled pins), a **pinch-off groove** around the cavity in the lower half so the closing mold squeezes excess resin into a land, kept away from the pod. If a groove outline can't be built the mold finishes without it and warns, rather than failing.
- **Plugs** for the wing socket and the motor pod, extracted from the part's own cavities so that hand-customised pod interiors are reproduced.
- **Bolt cones** in the stab mount, undersized and tapered for release, on one face only, to be drilled to size after cure.
- **Tiles** the halves to your printer's bed, with staggered joints and keys between tiles, and a keep-out band so joints never fall on the pad, nacelle or pod.
- **Layup** from the part's own volume and surface: woven skins, UD caps sized as a percentage of the core inside the skins over a percentage of the chord, tow for the rest, the mast plate's woven fill, and the mast's cap run-out and tab plies into the plate. It prints the per-cap cut list and the material and resin to buy.
- **Stiffness** from that layup against the loads Foil performance published: tip deflection and root cap stress for wings; for the mast, the side-load case with deflection at the fuselage end, cap stress at the plate and at its worst point along the height, track bolt tension, the junction shear check and the plate's bending.
- **Cut templates:** a 1 mm plate beside the mold with every fabric outline cut through it and a row of tally holes under each shape for its cut count. Export the plate's top face as DXF.

4. Reading the results

4.1 The performance notice

Cruise V km/h: L/D x · drag x N · x W | stall x km/h | take-off at x % of CL max | trim CG x mm · static margin x % MAC | root x Nm at n g, caps x mm

- **L/D, drag, power** at cruise. Power is hydrodynamic, drag times speed. For an eFoil, battery power is roughly drag power divided by 0.45 to 0.6 overall efficiency; the Propulsion group computes it properly.
- **Stall** is the speed below which the foil cannot fly level with the whole foil trimmed.
- **Take-off at x % of CL max**: over 85 % is marginal, since gusts, pumping and turning all add load; 100 % means it cannot fly at that speed.
- **Trim CG** is where the combined centre of mass must be, in millimetres aft of the wing pivot, for level flight. The rider places it with their stance. It is an output.
- **Static margin** is the distance from the trim CG forward of the neutral point as a percentage of MAC. Positive is pitch-stable. It is lowest at low speed, so check the take-off row, not just cruise. With a motor pod, judge the powered static margin instead.
- **Root and caps**: half-wing root moment at the load factor and the UD cap thickness needed. "Impossible" means the root section is too thin for that load.

4.2 The CSV

speed_kmh, alpha_deg, CL_wing, CL_stab, Re_wing_M, drag_N, drag_induced_N, drag_profile_N, L_D, power_W, trim_cg_mm, static_margin_pct, max_local_cl_pct, drag_mast_N, drag_body_N and, with propulsion, thrust_N, prop_rpm, J, eta_prop_pct, motor_A, throttle_pct, batt_A, batt_W, Wh_per_kmh, max_thrust_N, in_limits, thrust_couple_Nm, thrust_angle_deg, thrust_vert_N, trim_cg_powered_mm, static_margin_powered_pct.

alpha_deg is the board attitude relative to the flow. The derived quantities that matter most:

- **Best-L/D speed and minimum-power speed.** If best L/D sits well below the cruise you care about, the wing is too big or too cambered for that speed; if above, too small.
- **Drag split.** Induced above profile means the wing is working hard: more span or AR helps. Profile above induced means excess wetted area or thickness: less area, thinner sections or a smaller stab. They are equal at best L/D.
- **Stance shift,** trim CG at take-off minus trim CG at max speed. Large means the rider must move a lot as speed changes, strong speed stability; small means a neutral foil that holds any speed but corrects less.

4.3 Quick scaling rules

- Stall speed goes with $1/\sqrt{(\text{area} \times \text{CL max})}$: 20 % more area gives about 10 % lower stall speed.
- Lift at a given speed and angle goes with $\text{area} \times \text{speed}^2$.
- Induced drag at a given lift goes roughly with $1/\text{span}^2$, so at fixed area with $1/\text{AR}$. It dominates at low speed.
- Profile drag goes roughly with $\text{wetted area} \times \text{speed}^2$, weakly reduced by Reynolds number. It dominates at high speed.
- Tail volume = $\text{stab area} \times \text{arm} / (\text{wing area} \times \text{wing MAC})$.

- Power = drag × speed, so at high speed it grows roughly with speed³.

5. Design knowledge

5.1 Targets by discipline

Discipline	Area cm ²	Aspect ratio	Take-off km/h	Cruise km/h	Notes
Pump / downwind	900–1500	9–14+	8–12	12–18	Glide and low induced drag matter most; high AR, thin-ish sections
SUP / surf	1200–2500	4–7	10–14	14–22	Turning and low-speed lift; lower AR, more sweep and anhedral
Wing	900–1800	6–10	12–16	18–28	All-round compromise
Windfoil	700–1200	6–10	12–18	25–35	Higher speed; cavitation starts to matter
Kite race	300–700	8–12	15–25	30–45+	Small, thin, high speed; cavitation governs section choice
eFoil	1000–2000	5–9	12–16	18–25	Efficiency at cruise sets range; beginners want large area

Scale area roughly with total mass for the same take-off speed.

Section	Typical
Camber %	3–4 pump/SUP/surf, 2–3 wing/eFoil/wind, 1.5–2.5 kite; stab 0, or –1 to –2 inverted
Root thick %	11–14 front wing, 8–10 high speed, 8–11 stab
Section CL max (analytic)	1.0–1.2 for 8–10 % thick low camber, 1.2–1.4 for 12 % with 2–3 % camber, 1.4–1.6 for 14 % with 4 %; lower on small stabs

Stabiliser and pitch	Target	Why
Stab area / wing area	0.15–0.22	Below: weak pitch damping; above: extra drag
Tail volume	about 0.5–0.8	Higher is more stable and damped, slower to pitch
Arm	550–800 mm; 450–550 surf, 700–900+ pump	Longer is more stable, less pitchy
Geometric decalage	1.5–3°	Sets how much stance shifts with speed
Stab CL across the range	within about ±0.5; warning above 0.8	A stab near its own stall is unpredictable
Static margin at take-off	positive; under about 5 % MAC feels twitchy	The lowest point of the range
Stance shift take-off to max	tens of mm normal; over 100 mm is a lot	Depends on decalage and camber
Board attitude at cruise	within a few degrees of level	Riders dislike nose-up or nose-down

Performance sanity	Guide
Take-off % of CL max	under 80–85 %
Stall vs take-off	stall 15–25 % below intended take-off
L/D wing + stab only	high teens to mid 20s at best L/D; expect 5–10 points less at cruise once a real mast and fuselage are added
Cavitation number at max speed	should exceed the section's $-C_p$ min; below about 1.5 check with XFOil
Depth / MAC	above 2 to avoid free-surface effects

5.2 Levers: what each change does

Measured with the tool's own maths on a 1200 cm² AR 7 wing with a 200 cm² stab at 650 mm, 100 kg:

- **Decalage is the stance-shift and speed-stability knob.** More decalage (stab incidence more negative, stab camber negative, or wing incidence up) makes the stab push down harder at speed: the trim CG moves forward as speed rises, a large stance shift, a foil that resists speeding up, higher static margin, a little more drag at speed. Less decalage gives a neutral foil that holds any speed with little stance change, but low-speed static margin falls toward zero. Going from decalage 1.5° to 3° roughly doubled the stance shift from 12 to 30 km/h (75 to 118 mm); decalage 0 cut it to 33 mm.
- **Wing incidence does not change stall speed.** It changes board attitude almost one for one and acts like extra decalage. Use it to level the board at cruise, not to fix take-off.
- **Wing camber** lowers the attitude needed and raises the real CL max, but adds nose-down moment: bigger stance shift and worse high-speed drag. Pair more camber with slightly less decalage.
- **Stab area and arm** raise tail volume and static margin and increase stance shift. Stab area adds drag at speed, about one L/D point at 30 km/h for 50 % more stab.
- **Aspect ratio at fixed area** is the big efficiency lever at low and medium speed: AR 7 to 10 gave +3.7 L/D at 12 km/h and +2.5 at 20 km/h, nearly neutral at 30. Costs: a longer, more flexible wing with a higher root moment, slower roll, less turning.
- **Area** trades low-speed for high-speed performance.
- **Powered foils** add the thrust line as a pitch lever: see 5.5.

Symptom	First levers	Watch out for
Stalls at or near take-off	more area; more camber with a realistic CL max	top-end drag
Take-off fine but poor glide or pumping	more AR at the same area	root moment and cap thickness rise with span
Poor top speed, high power at speed	less area, thickness, camber; smaller stab	take-off speed rises
Best L/D well below cruise	wing too big or too cambered for that speed	
Best L/D well above cruise	wing too small: more area or AR	
Large stance shift	less decalage; less wing camber	low-speed static margin falls
Twitchy at low speed	more decalage, larger stab, longer arm	stance shift, drag
Nose-up or nose-down at cruise	wing incidence, and stab incidence by the same amount	
Stab CL near ± 0.8	larger stab, or stab incidence toward the unloading value	

Symptom	First levers	Watch out for
Cavitation warning	thinner, less cambered sections; more depth	low-speed lift
Caps too thick or impossible	thicker root, wider caps, lower span, lower load factor	drag, weight
Twitchy under power	lower Pod height; longer Arm; motor only from a higher speed; then stab area	Pod height also sets eFoil versus foil assist
Stands much further back under power	thrust line far above the drag centroid: lower Pod height	
High Wh/km, short range	ride near the best-Wh/km speed; less drag; higher P/D if throttle is low at cruise	static thrust falls with P/D
Top speed too low	higher P/D, more cells, lower gear ratio if pitch-limited, more current if current-limited	

5.3 Sections and polars

E817 and E818 are the common DIY hydrofoil sections; NACA 63-412 is a representative 6-series laminar section. E817 and E818 have camber peaking around 67 % chord, so their zero-lift angle and pitching moment are larger in magnitude than a NACA 4-digit of similar camber. That is real aerodynamics, not a fault, but trim and decalage numbers will differ when you switch. The most accurate combination is a tabulated profile with a section polar: the shape from the table, the CL max and drag bucket from a real polar run at the actual Reynolds number. On a polar wing the Section CL max field is ignored; a better polar is the fix, not a bigger number. The stall margin is evaluated at the take-off Reynolds number, interpolated between the polar's two Reynolds numbers; the warning tells you when it is extrapolating.

5.4 Pitch stability and the linked stab

Static margin is lowest at the lowest speed, and under power it is lowest at the lowest powered speed, usually take-off. On the foil-assist example the pod's thrust line is what sets the lowest margin: see 5.5 and 7.1. The stab's height relative to the wing changes the trim only slightly; on a tube fuselage it is dictated by the fuselage anyway, which is why Height is measured from the fuselage centreline.

5.5 Powered foils: propulsion, thrust line and thrust angle

Layout. Pod height decides what the foil is. If the pod can only clear the water with the wing shallower than two MAC, it is an eFoil and the motor is always in. If the wing can still ride two MAC deep with the pod out, it is foil assist, and both modes are reported at their own depths. Riding with the pod partly submerged is the worst case, friction plus spray on its waterline.

Method. At steady speed thrust equals drag. The B-series polynomials turn the drag at each speed into prop rpm, torque, shaft power and prop efficiency; motor current follows from torque and kV, motor volts from rpm and winding resistance, throttle from pack volts, then battery watts, Wh/km and range. Top speed is where full-power thrust, falling with speed, meets drag rising with speed squared. Static thrust should be several times cruise drag because ploughing before lift-off is not modelled.

Thrust couple. Thrust and drag form a pure couple, so the CG height does not matter. A thrust line above the drag centroid, typical of a high pod, is a nose-down couple: the rider stands further back under power and the powered static margin falls. The couple barely changes with speed while the stab's correcting moment grows with speed squared, so the lowest powered margin is at the lowest powered speed. Lowering the pod is the direct fix; a longer arm and then stab area come next.

Thrust angle. The thrust vector is fixed to the body, so as the board pitches up at low speed the thrust points partly upward: on the foil-assist example it makes 7.3° to the flow at 15 km/h and gives about 9 N of vertical thrust. Pod tilt lets you aim the pod nose-down so the thrust is aligned at cruise and helps at take-off; the console reports the angle, the vertical component and the couple at every speed.

Targets. Powered static margin at least 5 % MAC from the lowest powered speed; cruise throttle 40 to 70 %; cruise motor current a third to a half of the limit; foil-assist pod-out depth at least two MAC or it is not a real mode.

5.6 Mast shaping

- **Taper side.** A straight leading edge moves the plate, and so the board and rider, aft relative to the wing; a straight trailing edge moves them forward. The trim CG in the performance output tells you where the rider needs to be; the taper side is one way to put the board there.
- **Thickness where it counts.** The side-load moment on the mast peaks at the plate, and the top of the mast is dry at ride depth. On the foil-assist example only 340 of 750 mm is submerged at 350 mm depth, so the waterline sits at about 45 % of the height: thickening from 50 to 60 % adds stiffness with no wetted thickness and no drag. Starting the thickening very high bunches all the build-up plies into the top few centimetres, where they do little.
- **The worst stress is where the thickening starts, not always at the plate.** The moment grows all the way up the mast but the section only thickens from *Thicken from* %, so the cap stress peaks just below that height. On the foil-assist example it is 243 MPa at about 500 mm against 198 MPa at the plate. The mold's stiffness line reports both; start the thickening lower or raise *Thickness* to flatten the peak.
- **Cap thickness barely matters once the caps are thick:** thicker caps sit closer to the neutral axis. Section thickness and where the thickening starts are the levers.

5.7 The stab-to-fuselage joint

Studied from a real Axis rear wing (STL) and a builder's video: the stab carries a flat mount pad on top, faired up from the leading edge over about 45 % of the chord, 22 mm wide, standing about 7 mm off the section; the tail of the mount continues behind the trailing edge as a fairing that becomes the lower half of the fuselage's tail; two M6 bolts at 35 mm pitch pass up through the stab into the fuselage. The fuselage sits over the stab on a flat seat that matches the pad contour exactly, so there is no gap and no step. Molded bolt holes are made undersized and tapered so the mold releases, then drilled to size after cure, and only from one face so the top and bottom cones never meet. The tool reproduces all of this, pins the interface to the stab pivot so any stab fits any fuselage built for the same arm, and places the seat by its offset from the fuselage centreline. The same joint works upside down, with the stab on top of the fuselage and the pad underneath, which suits a foil-assist fuselage.

6. Building knowledge: layup, structure, molds and printing

6.1 The layup schedule

The mold sizes the layup from the part itself: its volume is the laminate volume, its surface carries the woven skins.

- **Skins:** *Skin plies* of *Skin gsm* woven on each face. On a two-ply skin turn the weave 45° on the outer ply for torsion.
- **UD caps:** *Cap thick %* of the *core inside the skins* (the local section less both skins), over *Cap width %* of the chord about the thickest point. The count is per cap; every UD template is cut twice, one per face. Ply k, outermost first, runs out to the last station where the stack is still k plies thick, so the caps taper along the span and the cut list gives every ply's length and its width at the root and at its end. On a mirrored wing a cap ply is one piece across the centre.
- **Tow:** whatever the skins and caps do not fill, given as a mass and a length at the chosen tex.
- **Cured ply thickness** = $\text{gsm} / (1780 \times \text{fibre volume fraction})$; 300 gsm UD at 50 % is 0.34 mm.
- **Overcharge %** adds material above the exact fill so the closing mold packs the laminate and squeezes flash into the pinch-off. Cloth and resin waste percentages turn the in-part figures into what to buy and mix. Fibre volume 45 to 50 % for a wet layup clamped in a matched mold, 55 to 60 % vacuum bagged or prepreg.
- **Mast:** the cut list runs from the plate down, so the thickened top adds short plies at the plate end; the plate is a solid block of woven fill between its skins, with its own cut count; every cap ply runs out over the root fillet onto the plate; tab plies interleave with the caps, running down the mast and out over the plate, stitching the two surfaces together through the stack.

6.2 Stiffness and strength

Per station, bending stiffness $EI = \text{UD modulus} \times \text{cap second moment} + \text{woven modulus} \times \text{skin second moment}$ (0.7 of the extreme fibre) + tow modulus $\times$ a lens-shaped core term. Curvature M/EI is integrated twice from the root to give the tip deflection; cap stress is $E \times M \times y / EI$ at the root. Wings use the spanwise moment Foil performance publishes at the load factor. On the foil-assist example the wing at 4 g deflects 6.5 mm at the tip with 137 MPa in the caps, and the stab 1.9 mm with 100 MPa.

Ply drops. Plotting the stress with the real, whole-number ply count at each station instead of the smooth taper shows the drops change the local stress by about 1 % where it matters. The wing is limited by its taper, not by where the plies stop, so lay the drops where convenient and stagger their ends.

Mast side load. The mast is fixed at the plate by the board and rider, and a side force of *Mast side g* times the all-up weight is applied at the fuselage end, so the moment is force $\times$ height and peaks at the plate. The tool reports the deflection at the fuselage end, the cap stress at the plate, the track bolt tension (moment / track spacing / bolts per row), the plate's bending stress between the tracks, and the junction check below. One g of the whole all-up mass sideways is a severe case; a hard carve is nearer 0.3 to 0.5 g.

The mast-to-plate junction is the weakest point. In a 20 mm section the two caps are only about a centimetre and a half apart, so each carries tens of kilonewtons at the plate; on the foil-assist example about 38 kN. That force has to leave the UD by shear into the plate. With the caps folded 47 mm over the plate and nothing else, the shear would be about 19 MPa against an allowable of 15 (interlaminar shear strength 30 MPa with a factor of 2): the caps would peel off the corner. Four interleaved tab plies per face give nine bonded interfaces and bring it to about 2 MPa. The root fillet gives the plies a radius to turn through instead of a sharp fold. Tab plies take their thickness out of the tow core over the tab length; the layup line warns if there is not enough core.

6.3 Mold design

- **Printed PLA molds, carbon parts.** The molds are two halves parted along the camber line's silhouette so both halves release, with a 20 mm base and a 25 mm flange by default.

- **Stations and tip bias.** The parting curtain is lofted through up to 80 stations. Sections with a curled tip (a 6-series section with a strong tip drop) failed at 40 evenly spaced stations and worked at 50 with a tip bias of 4, which packs the stations toward the tip. If a mold fails to loft, raise the station count and the bias first.
- **Keys and pins.** Printed frustum keys with 12° draft, or drilled pin holes for dowels; pin fit clearance around 0.2 mm.
- **Pinch-off.** A land and a groove around the cavity in the lower half so the closing mold squeezes excess resin out and packs the laminate; kept off the motor pod so the pod's plug is undisturbed.
- **Plugs.** The wing socket and the motor pod need cores. The mold extracts them from the part's own cavities, so a hand-modelled motor interior, placed above the mold in the feature list, is reproduced in the plug.
- **Bolt cones.** Undersized, tapered, on one face only, drilled to size after cure. Molded holes become pins in the mold cavity; two opposing pins cannot meet.
- **The mast mold** is three parts: the mast laid flat, split at mid-thickness into left and right, plus a tray that forms the plate top with straight locator pins. Countersunk heads in the tray were tried and abandoned because they would not release.
- **Tiling.** Halves larger than the printer bed are cut into tiles with staggered joints and keys, and joints are kept away from the stab pad, the wing nacelle and the mast pod.
- **Cut templates.** The template plate carries the woven skin outline developed along the tip droop, one tapered strip per UD ply, the mast's plate outline, the tab strips with fold notches, and a row of 3 mm tall holes under each shape for its cut count. Right-click the plate's top face in Onshape and export as DXF. A standalone script produces the same DXF from a pasted console log for a shop without Onshape.

6.4 Printing the molds

These are starting points, not proven settings: the example molds have not been printed yet. Tune them on a test tile before committing a whole mold.

Which material. The deciding number is heat. Epoxy gives off heat as it cures, and a thick laminate with a fast hardener can reach 50 to 70 °C in the core. PLA starts to soften at about 55 to 60 °C, PETG at about 75 to 80 °C. So:

- **PLA** is fine for the wing and stab molds with a slow hardener, cured at room temperature, and demolded before any post-cure. It prints easily, holds dimensions well and sands cleanly.
- **PETG** is the safer choice for the mast mold, whose 20 mm root and tow core make the most heat, and for any mold you want to post-cure in, up to about 60 °C. It is tougher and less brittle when you clamp it, but harder to print accurately and harder to sand.

Orientation: stand each tile on its end. Print each tile standing on its spanwise end, the flat cut face where it meets the next tile, with the span vertical. Every layer is then a slice through the mold that contains the whole section profile, so the nozzle draws the cavity curve smoothly in one pass per layer, leading edge to trailing edge. The only layer steps run along the span, where the surface changes slowly, and the lines they leave run chordwise with the flow. Printed flat, base down, the same curve would be built from stacked layers and every one would leave a step across the chord in the cavity.

It also helps elsewhere. The end face that sits on the bed is the joint face, so it comes out flat and square. The tile-to-tile keys point straight up and print cleanly. Clamping force acts within each layer rather than trying to peel them apart.

The costs: a tall, narrow print needs a good brim and steady adhesion. The half keys on the flange face now point sideways and need a little support, or can be printed separately and glued in. Each tile's span length must fit your printer's height. The tool sizes tiles for a base-down print: *Printer X* limits the chord, *Printer Y* the span length of each tile, and *Printer Z* the block height. Standing on end, the span length goes up the printer's height and the block height lies across the bed, so enter your printer's height as *Printer Y* and the bed's second dimension as *Printer Z*. On a printer with the same size in all three directions, such as 250 × 250 × 250, nothing changes: the example's 225 mm tiles fit either way.

Setting	PLA	PETG	Why
Nozzle	205-215 °C	235-245 °C	Good layer bonding; the mold is clamped and must not split
Bed	55-60 °C	75-85 °C	The end face on the bed is a joint face; it must stay flat
Cooling fan	100 % after the first layers	30-50 %	PETG bonds better and strings less with less cooling
Speed	50-80 mm/s, outer walls 30-40	40-60 mm/s, outer walls 25-30	The cavity is an outer wall in every layer; slow walls give a smoother surface to fair
Layer height	0.16-0.20 mm	same	Steps now run along the span, where the surface is gentle; thinner layers only help at a strongly drooped tip
Walls	5 perimeters (at least 2 mm)	same	The cavity is a wall now; there must be solid material under the sanding
Top and bottom	4-5 layers	same	Only the tile's two end faces are top and bottom surfaces
Infill	15-20 % adaptive cubic (or gyroid); 40 % only under the flanges and keys, with a modifier	same	The walls make the working surface; the core only carries clamp load to the base. Dense gyroid through the whole block more than doubled the print time on a test slice
Seam	aligned to a back corner of the base, never in the cavity	same	Every layer crosses the cavity, so a random seam would put a dot in it every layer
Brim	8-10 mm, essential	8-10 mm, essential	Tall, narrow tiles on a small end face
Supports	only under the flange half keys, if printed in place	same	Everything else is self-supporting
Elephant's foot compensation	0.1-0.2 mm	0.15-0.25 mm	The bed face is a joint face and must be square to seat the next tile

What it costs. The stab mold for the foil-assist example is five tiles on one 250 mm bed, with the tool's default 25 mm flange and 20 mm base. It sliced as follows:

Settings	Filament	Time
4 walls, light infill	1.17 kg	16 h 38 m
6 walls, dense gyroid infill	1.69 kg	39 h 14 m
5 walls, light infill (recommended)	1.29 kg	19 h 21 m

The dense infill more than doubled the time for almost no benefit; the fifth wall costs about an hour and a half and gives the cavity material to be sanded. For a stab, the tool's block can also shrink: *Flange width* 18

and *Base thickness* 16 still clear the key limits with 12 mm pins and 14 mm keys, and take about a quarter of the plastic off. Print a stab mold in two batches, upper tiles then lower, if one failure costing the whole bed would hurt.

Fit. The tool's *Pin fit* (0.2 mm by default) sets the clearance between each key and its socket. Print one key and socket pair first; if it binds, raise *Pin fit* by 0.05 mm and regenerate rather than sanding every key. Shrinkage over a 250 mm tile is well under half a millimetre, and the keys take it up; you don't need to scale the model.

Finishing before the first layup.

1. Glue the tiles together on their keys with epoxy or CA, clamped on a flat surface.
2. Sand the cavity and parting faces, working from about 120 to 400 grit, until the layer lines are gone.
3. Seal the surface: two or three coats of laminating epoxy, or a high-build epoxy primer, sanded between coats. Raw PLA and PETG are porous and epoxy will bond to them.
4. Wet sand the sealed surface to 800-1200 grit. The mold finish is the part's finish.
5. Release: five or more coats of mold release wax, buffed between coats, then optionally a PVA film release. Re-wax before each layup.
6. Avoid solvents on the mold: acetone attacks PLA and softens PETG.

7. Worked example

7.1 A prone wave foil assist, 95 kg all-up

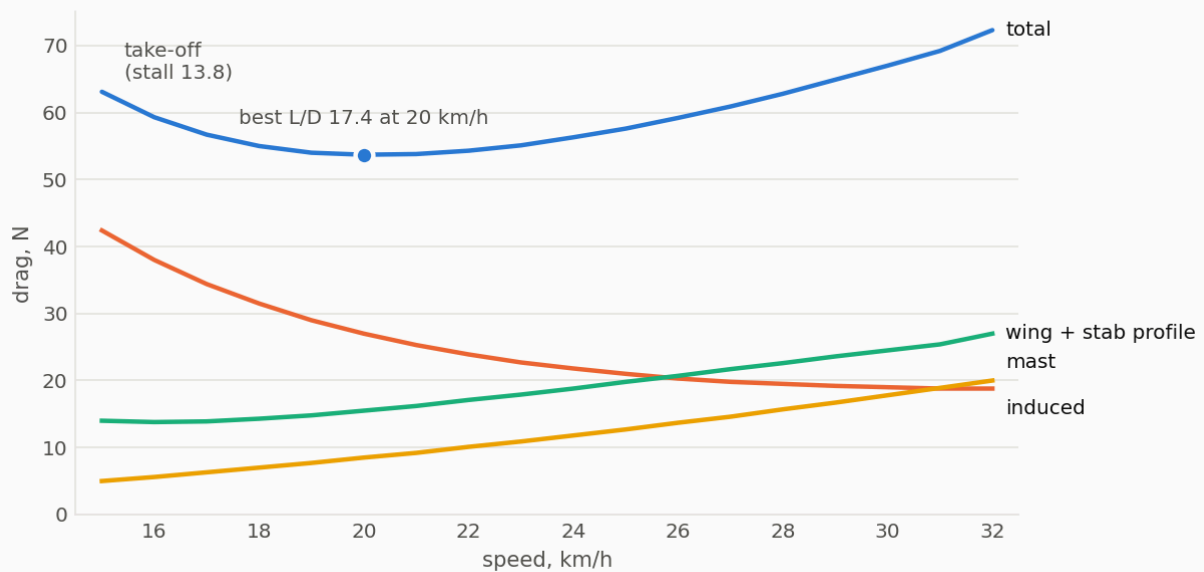
A foil built to test the theory. More examples, built from published parameters of real foils, may follow. Rider 80 kg, board and kit 15 kg. Wing 1000 cm², AR 7.2, 850 mm span, NACA 4-digit with 3 % camber, 12/10 % thick, 100 mm tip sweep, 30 mm tip drop, 1° washout, 1.5° incidence, 1 mm minimum tip. Stab linked at 18 % (180 cm², AR 5.7, symmetric 10/8 %) at -1.5°, 660 mm behind the wing, its seat 4 mm below the fuselage centreline (*Height* -4), two M5 bolts at 35 mm. Mast 750 mm, 100 to 110 mm chord, 16 mm thick rising to 22 mm at the plate from 60 % height, 12 mm root fillet, 63 mm pod at 550 mm so it is out of the water once foiling. 24 mm tube fuselage. Drive: 160 mm three-blade prop, P/D 0.8, 140 kV direct on 10S, 120 A, 20 Ah.

Pod out, steady state:

km/h	Board angle	CL wing	CL stab	Drag N (induced / profile / mast / fuselage)	L/D	W to hold speed
15	7.4°	1.04	+0.05	63.1 (42.4 / 14.0 / 5.0 / 1.6)	14.8	263
18	4.1°	0.74	-0.09	55.0 (31.5 / 14.3 / 7.0 / 2.3)	16.9	275
20	2.7°	0.62	-0.15	53.7 (27.0 / 15.5 / 8.5 / 2.8)	17.4	298
25	0.4°	0.42	-0.25	57.6 (21.0 / 19.8 / 12.7 / 4.1)	16.2	400
30	-0.8°	0.32	-0.30	67.0 (19.0 / 24.5 / 17.8 / 5.8)	13.9	558

Where the drag comes from

Prone foil assist, 95 kg all-up, 1000 cm² wing, pod out, steady state



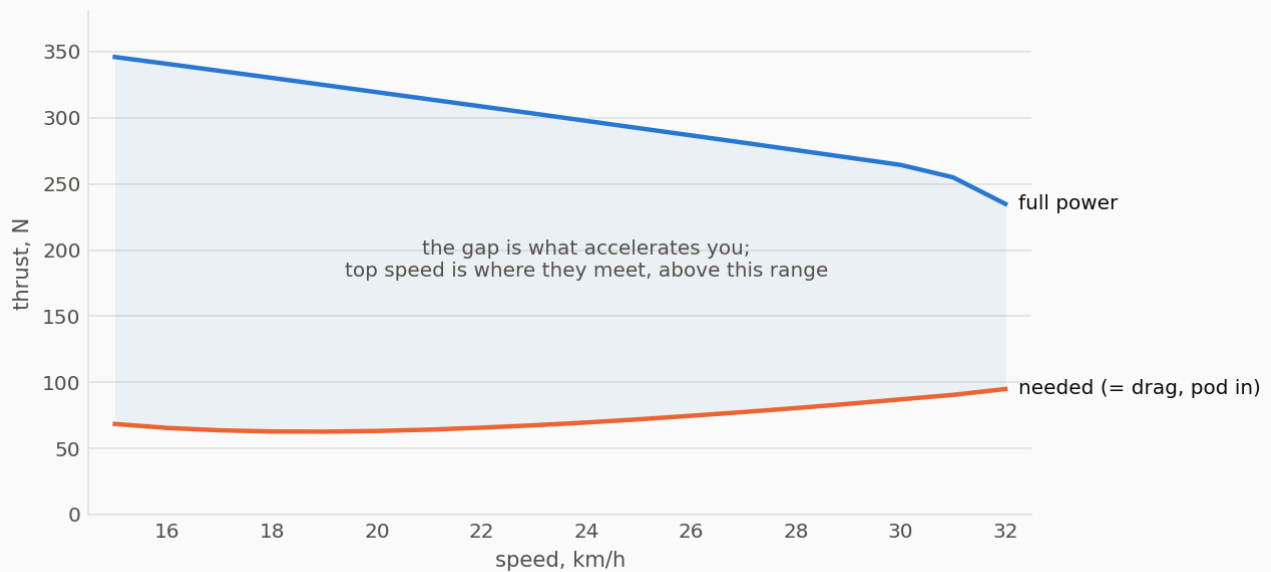
The model's stall is 13.8 km/h, with take-off at 15 at 86 % of the section's CL max. An experienced builder's estimate for a real foil like this is nearer 18 km/h, which corresponds to an effective CL max of about 0.75 against the section's 1.27: see 10.1 before trusting a stall speed. Best L/D 17.4 at 20 km/h. With the pod in the water, drag rises about 7 N and L/D falls to 14.7.

Motor on, pod in:

km/h	Thrust N	rpm	Prop eff.	Motor A	Throttle	Battery W	Wh/km
15	68.5	2160	69 %	31	46 %	539	36
20	63.2	2600	64 %	34	55 %	718	36
25	72.0	3140	59 %	44	67 %	1116	45
30	87.1	3710	55 %	57	79 %	1730	58

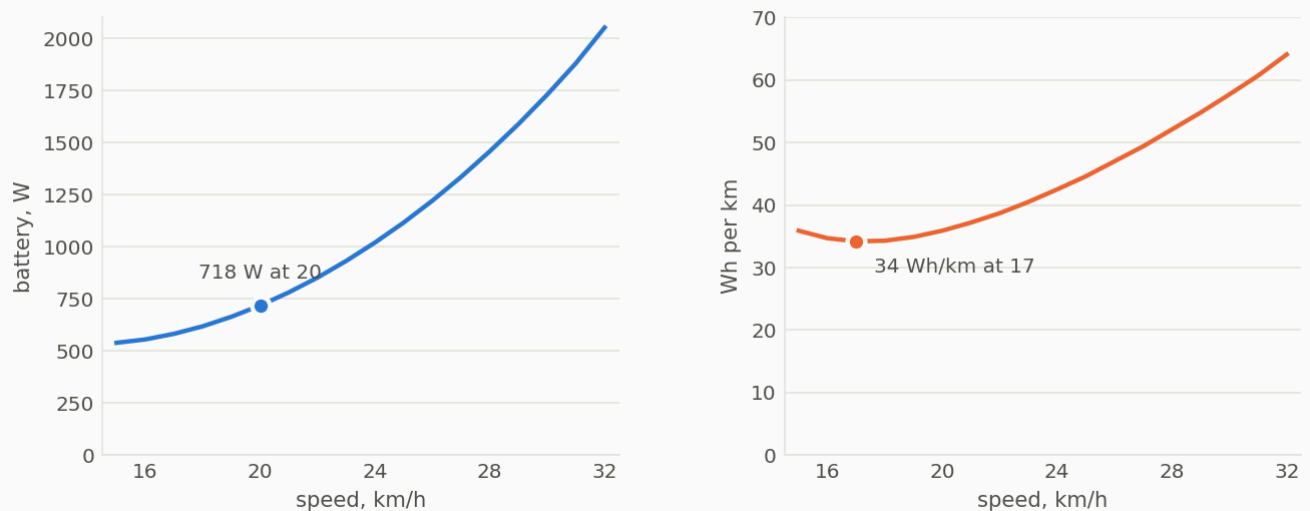
Thrust available against thrust needed

Motor on, pod in: 160 mm prop, P/D 0.8, 140 kV on 10S, 120 A limit



What the pack pays

Motor on, pod in. Left: battery watts. Right: watt-hours per kilometre, lowest around 17-20 km/h

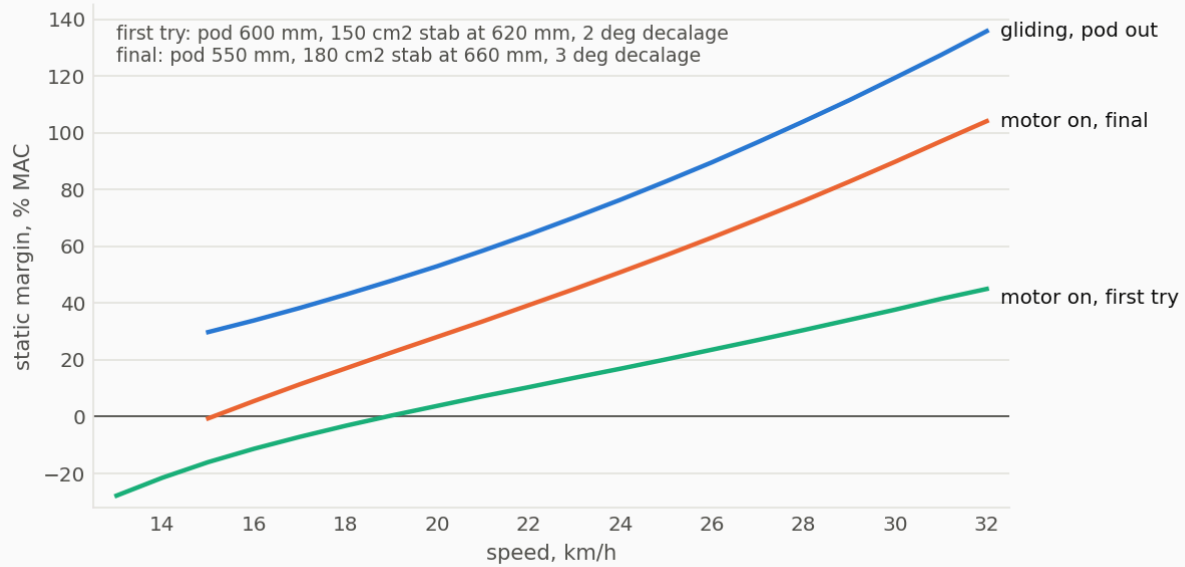


Static thrust 41 kgf, top speed 37 km/h, 18.5 km or about 56 minutes of continuous powered cruising.

Pitch. Gliding, the static margin is 30 % of MAC at take-off and 53 % at cruise. Under power the thrust line is 560 mm above the wing and 450 mm above the drag centroid, a 34 Nm nose-down couple at take-off, and the powered margin at 15 km/h is -0.6% , rising to $+5.5\%$ at 16: neutral at take-off. The first version, with the pod 50 mm higher, a 150 cm^2 stab at 620 mm and 2° decalage, read -16% . Foil assist trades powered stability for a dry pod, and pod height is the lever.

Pitch stability with the motor on and off

Static margin, % of wing MAC. Positive is stable; the lowest point is at take-off



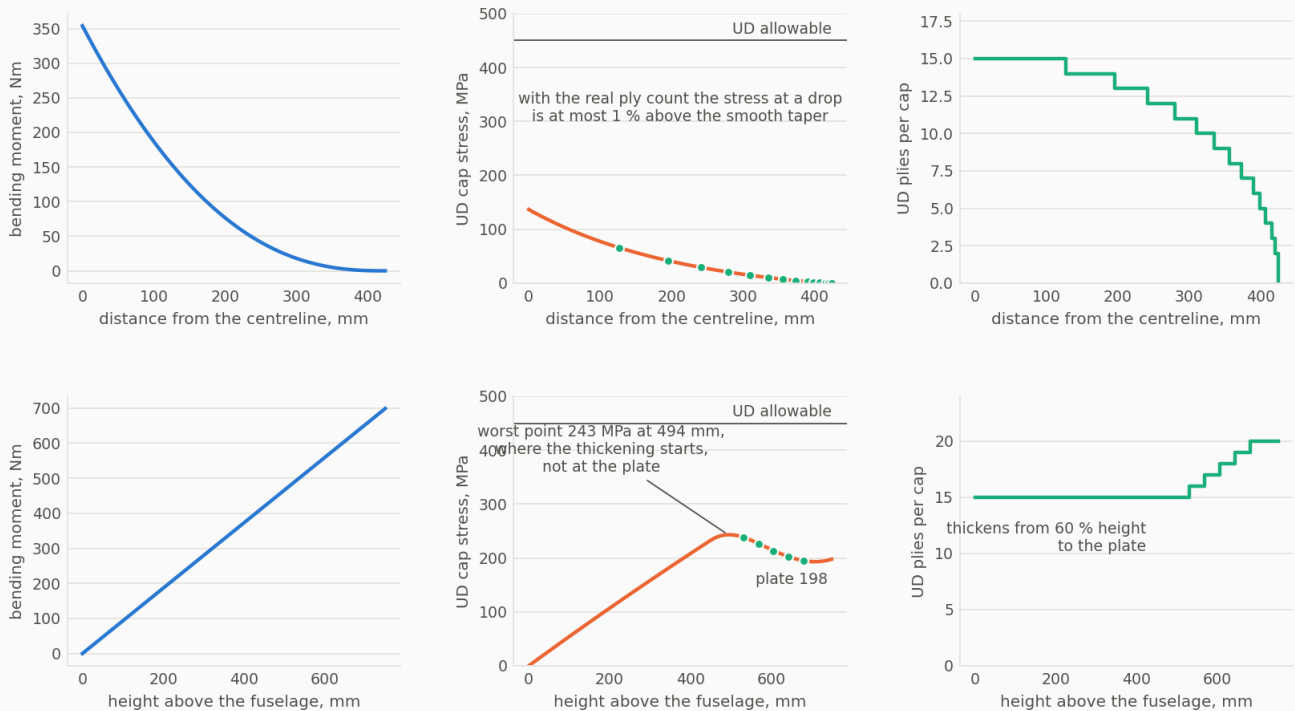
Structure, from the layup the mold feature wrote:

Part	Load case	Layup	Deflection	Cap stress
Wing	4 g, 354 Nm at the root	2 × 200 g woven each face, 15 UD plies per cap tapering to 4, tow core	6.5 mm at the tip	137 MPa
Stab	4 g	same skins, 5 UD per cap	1.9 mm	100 MPa
Mast	1 g sideways at the fuselage, 932 N	20 UD per cap at the plate, 15 at the foot, 4 tabs per face into the plate	53 mm at the fuselage end	198 MPa at the plate, 243 MPa at 488 mm

Mast junction: 38 kN cap force, 2.1 MPa shear against 15 allowed, 3.9 kN per track bolt, 96 MPa in the plate.

Loads and stresses, from the layup the mold feature wrote

Top: wing at 4 g, 354 Nm at the root. Bottom: mast with 1 g sideways at the fuselage end, 932 N, plate fixed. Dots mark ply drops



Materials at 50 % fibre volume: wing 1.6 kg cured (0.43 m² woven, 1 m² UD, 600 g tow, 830 g resin mixed), stab 0.13 kg, mast with plate 1.9 kg; about 3.7 kg of carbon and epoxy for the three molded parts. Wing mold 225 × 899 × 91 mm in nine tiles for a 250 mm bed, stab mold in two halves with the bolt cones, mast mold in three.

8. Parameter reference

Every dialog field of every feature, with its limits, default and the tooltip text, generated from the Feature Studio itself. Fields marked as appearing only when a checkbox or dropdown enables them are hidden otherwise.

Generated by `tools/gen_parameters.py` from the Feature Studio (build 141; wing 1.020, stab 4.002, perf 1.013, mast 2.008, fuselage 4.012, mold 2.069). Do not edit by hand - re-run the script when the .fs changes.

Exact dialog labels, limits and defaults (min / default / max). Use these labels when recommending a change. The meaning column is the Onshape tooltip. Some fields only appear when a checkbox or dropdown above them enables them.

Foil wing

Section

Label	Min / default / max	Meaning
Profile	dropdown: NACA 4-digit, Eppler E817, Eppler E818, NACA 63-412 (default NACA 4-digit)	Section family. NACA 4-digit is fully parametric (camber, camber position, thickness) - use it for anything except matching a known published section. The Eppler and NACA 63-412 options are real digitised coordinate tables (UIUC Airfoil Coordinates Database), thickness-scaled to your Root/Tip thick % (Camber % is fixed by the table and hidden). Custom .dat paste is a later phase.
Camber %	0 / 3 / 9	Maximum camber as % of chord. Limits 0-9 %. 0 = symmetric. Typical front wing: 3-4 % pump/SUP/surf (early lift), 2-3 % wind/eFoil, 1.5-2.5 % kite/high speed. More camber = more lift at low speed but more drag and a bigger nose-down pitching moment. Only shown for NACA 4-digit - the tabulated profiles carry their own fixed camber.
Camber pos %	10 / 40 / 90	Position of maximum camber as % of chord from the leading edge. Limits 10-90 %. Typical 30-50 %, 40 % is the classic NACA choice. Further forward = gentler stall; further aft = more aft loading and a stronger nose-down moment. Ignored when camber is 0. Only shown for NACA 4-digit.
Root thick %	4 / 12 / 25	Root section thickness as % of chord. Limits 4-25 %. Typical front wing 11-14 %, high-speed/kite 8-10 %. Thicker = stiffer, more room for spar caps and a more forgiving stall; thinner = less drag and later cavitation.
Tip thick %	4 / 10 / 25	Tip section thickness as % of chord. Limits 4-25 %. Typical 8-11 %, usually 1-3 points thinner than the root. Blend between root and tip is set by the thickness exponent.
TE thickness	0.1 / 0.8 / 5 mm	Limits 0.1-5 mm, and at most 3 % of the root chord. Held constant along the span, then thinned automatically near the tip so it never exceeds 3 % of the local chord. Typical 0.6-1.0 mm for wet-laid carbon in printed molds (room for 2-3 plies to wrap the edge), 0.3-0.5 mm for prepreg in machined molds. Thicker = tougher edge but more base drag and a risk of humming (vortex shedding).
Points/surface	10 / 40 / 120	Limits 10-120. Typical 30-50. Cosine-spaced, so the leading edge is well resolved even at low counts. Above about 60 slows regeneration for no visible gain.

Planform

Label	Min / default / max	Meaning
Planform input	dropdown: Area + aspect ratio, Span + root chord (default Area + aspect ratio)	Area + aspect ratio (recommended): enter the area everyone quotes and the shape; span and root chord are solved. Span + root chord: enter the geometry directly (e.g. to copy an existing foil); area becomes an output.
Area cm ²	100 / 1100 / 5000	Target wing area in cm ² . Limits 100-5000 cm ² . Typical front wing: 1500-2500 SUP/surf/beginner eFoil, 1000-1600 intermediate eFoil/wind, 900-1500 pump/downwind (high aspect), 300-700 kite race.
Aspect ratio	2 / 9 / 20	Limits 2-20, projected span ² / projected area. Typical front wing: 4-6 surf/beginner, 6-9 all-round/eFoil/wind, 9-14+ pump/downwind, 8-12 kite race. Higher = less induced drag and better glide, but less roll agility and more root bending.

Label	Min / default / max	Meaning
Area definition	dropdown: Projected area, Developed area (default Projected area)	Which area the target refers to. Projected (top view) is what sets lift and what most brands quote. Developed is measured along the curved span line and is larger with anhedral or gull. Both are reported.
Span	100 / 800 / 3000 mm	Limits 100-3000 mm, projected tip to tip (top view). Typical front wing: 700-950 mm SUP/surf/eFoil, 900-1300 mm pump/downwind, 700-1000 mm wind, 600-900 mm kite race.
Root chord	20 / 180 / 600 mm	Limits 20-600 mm. Typical front wing 140-250 mm (low/medium aspect) or 90-140 mm (high aspect, kite).
Chord distribution	dropdown: Superellipse, Linear taper (default Superellipse)	Superellipse: $c = c_0 (1 - (s^2 y/b)^n)^{1/n}$, cut off at the end chord ratio, giving a rounded tip. Linear: straight taper to a square tip.
Exponent n	1.2 / 3 / 8	Limits 1.2-8. 2 = elliptic (lowest induced drag for an untwisted wing). 2.5-4 = fuller, squarer tips as on many surf/eFoil wings. Below 2 = pointier, glider-like tips. Above 5 approaches a rectangle with rounded corners. Typical 2-3 for most wings, 3-4 for surf/eFoil.
End ratio	0.02 / 0.05 / 0.6	Limits 0.02-0.6. Chord of the small end face where the superellipse is cut off, as a fraction of root chord. The leading and trailing edges follow the curve all the way to this chord, so the tip is rounded in plan view. Typical 0.04-0.08 for a fully rounded tip; 0.15-0.3 gives a curved but blunt tip. The trailing edge is thinned automatically near the tip.
Tip ratio	0.1 / 0.4 / 1	Limits 0.1-1.0. Tip chord of the straight taper as a fraction of root chord; the tip is square in plan view. Typical 0.3-0.6; about 0.4 gives near-elliptic loading for an untwisted wing. Below 0.3 the tips load up and stall first.
Min tip	checkbox, default off	On = the tip chord is widened automatically until the section at the tip cap start is at least Min tip mm thick, by raising End ratio (or Tip ratio) above the value entered. The notice reports the ratio used. Guards a thin, small-chord tip that cannot be laid up; especially useful on a stab linked to a wing, where the link sets area, aspect ratio and thickness but the tip chord is still this feature's own.
Min tip mm	0.2 / 1 / 5 mm	Minimum section thickness at the tip cap start (the last full section before the cap tapers out). Limits 0.2-5 mm. Typical 1-1.5 mm for a carbon layup: two skin plies plus a little UD; the cap beyond it is tow and resin. Only the tip chord is changed, never the thickness %.
Stations	3 / 9 / 30	Limits 3-30, per half span. Typical 11-15 for a rounded superellipse tip, 5-9 for linear taper. Superellipse stations are spaced evenly around the tip curve; linear taper uses sine spacing. Tip-cap stations are added on top. More = slower regeneration.
Mirror to full span	checkbox, default on	On = complete foil. Off = right half only, for faster edits and section inspection. Reported metrics are always full-span values.

Sweep, alignment and anhedral

Label	Min / default / max	Meaning
Align line %	0 / 25 / 100	Limits 0-100 %. Typical 25 (classic elliptical), 60-100 for modern pump/downwind and most stabs. The chord line that follows the sweep curve; chord is added around it. 100 = straight trailing edge with the leading edge curving back (common on pump, downwind and many surf wings). 25 = symmetric look, both edges curve. 0 = straight leading edge. Independent of the incidence pivot. Moving it aft sweeps the quarter-chord line back at the tips, moving the lift centre aft.

Label	Min / default / max	Meaning
Sweep aft	-100 / 100 / 400 mm	Limits -100 to +400 mm, positive = aft. How far the alignment line sits aft at the tip, relative to the root. 0 = alignment line straight across. Typical 20-80 mm for all-round wings, 80-200 mm for surf/carving wings with a curved leading edge. More sweep = more yaw/pitch stability and forgiveness, less efficiency.
Sweep exp	0.5 / 2 / 4	Limits 0.5-4. Shape of the sweep line $x = \text{offset} (2y/b)^m$. 1 = straight swept line (kinked at the root), 2-3 = curved 'banana' leading edge that is tangent at the root, below 1 = sweep concentrated near the root, 3-4 = raked tips. Typical 2-3.
Tip drop	-100 / 50 / 200 mm	Limits -100 to +200 mm, positive = down (anhedral). Vertical drop of the tip pivot below the root pivot (net, including any gull dihedral); negative = tips up. Typical 0-40 mm for all-round wings, 30-80 mm for surf wings. Anhedral loosens roll and helps carving; too much reduces effective span and can create mold undercuts.
Drop exp	1 / 2 / 6	Limits 1-6. Typical 2-3. Shape of the drop $z = -\text{drop} (2y/b)^e$. 1 = straight anhedral from the root, 2-3 = flat centre with tips falling away, higher = droop only near the tips.
Gull dihedral	-10 / 0 / 20 deg	Limits -10 to +20 deg, positive = up. Slope of the span line leaving the root, for gull wings: the wing rises inboard, then the drop term pulls the tips down to the tip drop. Needs a drop exponent above 1 (no effect at 1). Typical 0 (plain anhedral) or 3-8 deg with tip drop 20-60 mm and exponent 2.5-4 for a gull. Creates a V at the root; large values can cause mold undercuts.

Twist and thickness distribution

Label	Min / default / max	Meaning
Washout	-5 / 0 / 8 deg	Limits -5 to +8 deg, relative to the root incidence; positive = tip leading edge down. Typical 0-3 deg. Washout unloads the tips so the root stalls first (gentler stall, better tip-breach behaviour) at a small cost in induced drag. Negative = wash-in (rarely useful).
Washout exp	0.5 / 1 / 4	Limits 0.5-4. Twist follows washout $(2y/b)^w$. 1 = linear, 2-3 = twist concentrated towards the tips, below 1 = twist concentrated near the root. Typical 1-2.
Thickness exp	0.5 / 1.5 / 4	Limits 0.5-4. Thickness % follows root + (tip - root) $(2y/b)^p$. 1 = linear, 2-3 = keeps the root thickness (and stiffness) further out before thinning, below 1 = thins quickly away from the root. Typical 1-2.

Incidence

Label	Min / default / max	Meaning
Incidence	-10 / 0 / 10 deg	Limits -10 to +10 deg, measured from the fuselage axis (X), positive = leading edge up. Typical front wing +0.5 to +2.5 deg. Aim for a wing-minus-stab difference (decalage) of 1.5-3 deg; the stabiliser feature reports it.
Pivot	dropdown: Quarter chord, Leading edge, Trailing edge (default Quarter chord)	Point each section rotates about for incidence and washout, and the line that follows the drop curve. The root pivot sits at the world origin. Quarter chord (recommended) keeps the lift centre fixed as angles change. Plan-view shape is set by the alignment line, not the pivot.

Tip

Label	Min / default / max	Meaning
Tip cap	dropdown: Rounded tip, Flat tip (default Rounded tip)	Rounded tip: section thickness falls along a quarter-ellipse over the cap length, rounding the tip in front view. Plan-view rounding comes from the superellipse end ratio. Flat tip: thickness is unchanged to the end face.
Cap length	2 / 15 / 80 mm	Limits 2-80 mm, and at most 30 % of the half span. Typical 10-25 mm. Longer looks smoother but thins more of the working span.
Cap end %	5 / 25 / 80	End-face thickness as % of the tip section thickness. Limits 5-80 %. Typical 15-30 %. Lower = rounder, thinner tip; very low values with a thick trailing edge give a wafer-thin end that is hard to lay up.

Section polar

Label	Min / default / max	Meaning
Section data	dropdown: Analytic, Polar summary, Built-in (default Analytic)	Built-in: fitted XFOil data, nothing to type in. E817/E818/NACA 63-412 use one real polar each. NACA 4-digit uses the exact zero-lift angle/moment for your Camber %/Camber pos %, plus lift slope, CLmax and drag bucket fitted from real NACA polars at your thickness/camber - coarser than the fixed profiles (a fit across a family, not one measured shape); use Polar summary for higher accuracy at your exact section. Analytic (default): thin-aerofoil model - camber-line angle/moment, fixed lift slope, skin-friction drag, Section CL max from Foil performance. Polar summary: your own XFOil/XFLR5 numbers at your own Reynolds numbers, via the fields below. Run scripts/polar_summary.py (developer skill) on your polar - the tool never invents section data.
Zero-lift	-8 / -2 / 4 deg	Zero-lift angle of attack from the polar, deg (usually negative for a cambered section). Limits -8 to 4 deg. Replaces the thin-aerofoil estimate. Typical -1 to -3 deg for a few % camber. For the built-in profiles (E817, E818, NACA 63-412) use Section data = Built-in instead of typing these in.
Lift slope	4 / 6 / 7.5	Section lift-curve slope from the polar, per radian. Limits 4-7.5. Thin-aerofoil theory gives $2\pi = 6.28$. Replaces the fixed analytic value. Typical 5.5-6.5 depending on thickness and Reynolds number. For the built-in profiles (E817, E818, NACA 63-412) use Section data = Built-in instead of typing these in.
Cm c/4	-0.15 / -0.05 / 0.05	Section pitching moment about the quarter chord from the polar (usually fairly constant with CL below stall). Limits -0.15 to 0.05. Replaces the thin-aerofoil Cm0. Typical -0.03 to -0.11 for a cambered section, near 0 for symmetric. For the built-in profiles (E817, E818, NACA 63-412) use Section data = Built-in instead of typing these in.
Re 1 (M)	0.1 / 0.35 / 5	First Reynolds number the polar data below is from, in millions. Limits 0.1-5. Values are interpolated (log Re) between Re 1 and Re 2, and held flat outside that range. Typical: near your take-off Reynolds number (about 0.2-0.5 M for a wing, lower for a stab). For the built-in profiles (E817, E818, NACA 63-412) use Section data = Built-in instead of typing these in.
CLmax @Re1	0.6 / 1 / 1.8	Maximum section lift coefficient at Re 1. Limits 0.6-1.8. Real CLmax usually drops noticeably below the higher-Re value, which is why a single fixed CL max (the analytic fallback) can be optimistic at take-off. Typical 0.85-1.1 at low Re. For the built-in profiles (E817, E818, NACA 63-412) use Section data = Built-in instead of typing these in.
Cdmin @Re1	0.004 / 0.01 / 0.03	Minimum section drag coefficient at Re 1 (the bottom of the drag bucket). Limits 0.004-0.03. Friction drag rises as Re falls. Typical 0.008-0.018 at low Re. For the built-in profiles (E817, E818, NACA 63-412) use Section data = Built-in instead of typing these in.

Label	Min / default / max	Meaning
CL@Cadmin 1	-0.3 / 0.3 / 1	The lift coefficient at which Cadmin occurs at Re 1 - the centre of the drag bucket. Limits -0.3 to 1.0. Typical 0.3-0.7. For the built-in profiles (E817, E818, NACA 63-412) use Section data = Built-in instead of typing these in.
k @Re1	0.005 / 0.013 / 0.05	Drag-rise rate at Re 1: $Cd = C_{admin} + k (CL - CL@C_{admin})^2$. Limits 0.005-0.05. For your own polar, read it off two points on the drag bucket away from Cadmin and solve for k, or use polar_summary.py to fit it automatically. Typical 0.005-0.04. For the built-in profiles (E817, E818, NACA 63-412) use Section data = Built-in instead of typing these in.
Re 2 (M)	0.1 / 0.9 / 5	Second Reynolds number, in millions. Limits 0.1-5. Must differ from Re 1 for interpolation to do anything. Typical: near your cruise or max-speed Reynolds number. For the built-in profiles (E817, E818, NACA 63-412) use Section data = Built-in instead of typing these in.
CLmax @Re2	0.6 / 1.2 / 1.8	Maximum section lift coefficient at Re 2. Limits 0.6-1.8. Typical 1.0-1.3 at higher Re. For the built-in profiles (E817, E818, NACA 63-412) use Section data = Built-in instead of typing these in.
Cadmin @Re2	0.004 / 0.007 / 0.03	Minimum section drag coefficient at Re 2. Limits 0.004-0.03. Typical 0.004-0.008 at higher Re. For the built-in profiles (E817, E818, NACA 63-412) use Section data = Built-in instead of typing these in.
CL@Cadmin 2	-0.3 / 0.3 / 1	The lift coefficient at which Cadmin occurs at Re 2. Limits -0.3 to 1.0. Typically close to CL@Cadmin 1, since the drag bucket centre moves less with Re than CLmax does. Typical 0.3-0.4. For the built-in profiles (E817, E818, NACA 63-412) use Section data = Built-in instead of typing these in.
k @Re2	0.005 / 0.01 / 0.05	Drag-rise rate at Re 2. Limits 0.005-0.05. Typical 0.01-0.035. For the built-in profiles (E817, E818, NACA 63-412) use Section data = Built-in instead of typing these in.

Tube socket

Label	Min / default / max	Meaning
Tube socket	checkbox, default on	Adds a low-drag tube socket to the wing root. It starts hidden inside the wing at the root's thickest point, blends smoothly out of the upper and lower surfaces to a round nacelle, and continues behind the trailing edge to a rear face where the round fuselage tube enters. The tube axis runs through the middle of the root thickness. Turn off for a clean wing (molding, analysis, or a different mount).
Tube dia	10 / 24 / 60 mm	Outside diameter of the round fuselage tube. Limits 10-60 mm. Typical 20-30 mm for carbon or aluminium tube.
Clearance	0 / 0.2 / 2 mm	Diametral clearance between bore and tube. Limits 0-2 mm. Typical 0.1-0.3 mm for a firm slip fit that is then pinned or bonded.
Wall	2 / 5 / 15 mm	Nacelle wall thickness around the bore, which sets the nacelle diameter (Tube dia + 2 x Wall). Limits 2-15 mm. Typical 4-6 mm for carbon. Thicker is stronger but adds frontal area and drag.
Blend length	10 / 70 / 300 mm	Length over which the socket grows out of the wing, from the root's thickest point (where it starts hidden inside the wing) to the full nacelle diameter. Limits 10-300 mm. Typical 50-90 mm - longer is a gentler, lower-drag blend. Should reach full diameter before or just after the trailing edge.
Rear extension	0 / 90 / 400 mm	How far the socket extends behind the root trailing edge to its rear face, where the tube enters. Limits 0-400 mm. Typical 60-120 mm; with Socket depth this sets how much tube is held and how far the joint sits behind the wing.

Label	Min / default / max	Meaning
Rear fairing	5 / 25 / 150 mm	Taper at the back of the socket, from the full nacelle diameter down to just over the tube (Tube dia + Wall) at the rear face - this is what keeps the fuselage slender: Foil fuselage matches the diameter this taper actually produces, not the nacelle's peak diameter. Limits 5-150 mm. Typical 20-40 mm; longer is gentler and lower drag. 0 is refused, since it would force the whole fuselage to the wide nacelle diameter.
Socket depth	20 / 80 / 300 mm	How far the bore runs forward from the rear face - the length of tube inside the wing. Limits 20-300 mm. Typical 3-4 x Tube dia; longer carries bending better.
Pins	0 / 2 / 3	Vertical pin or bolt holes through the nacelle and tube, spread along the socket. Limits 0-3. Typical 1-2. 0 = no holes (bonded tube).
Pin bolt	dropdown: M4, M5, M6, M8, M10, Custom hole sizes (default M6)	The cross bolt (or pin) through the nacelle and the tube: sets the clearance hole here and in the Foil fuselage's matching holes (M5 5.5, M6 6.5, M8 8.5 mm, 0.1 mm looser than a close fit so the two parts line up). Typical M6 for a 24 mm tube; keep it well under half the tube diameter. Custom hole sizes: type the hole below (a plain dowel, for instance).
Pin dia	3 / 6.5 / 12 mm	Pin or bolt clearance hole diameter (Custom only). Limits 3-12 mm. Typical 6.5 mm (M6) for a 24 mm tube; keep well under half the tube diameter.
Pin pitch	10 / 35 / 150 mm	Spacing between pins along X, centred on the socket. Limits 10-150 mm. Typical 30-40 mm. Ignored with one pin.
Log inputs	checkbox, default on	Prints every parameter on this feature to the FeatureScript console as one line, so the whole set can be pasted elsewhere (e.g. back to an AI assistant) along with the notice or CSV.

Foil stabiliser

Section

Label	Min / default / max	Meaning
Profile	dropdown: NACA 4-digit, Eppler E817, Eppler E818, NACA 63-412 (default NACA 4-digit)	Section family. NACA 4-digit is fully parametric (camber, camber position, thickness) - use it for anything except matching a known published section. The Eppler and NACA 63-412 options are real digitised coordinate tables (UIUC Airfoil Coordinates Database), thickness-scaled to your Root/Tip thick % (Camber % is fixed by the table and hidden). Custom .dat paste is a later phase.
Camber %	-6 / 0 / 6	Maximum camber as % of chord. Limits -6 to +6 %. Typical 0 for stabs. 0 = symmetric (most common stab section). Negative = inverted camber that lifts downward, trimming with less negative incidence; typical -1 to -2 % if used. Positive only for lifting stabs (rare). Only shown for NACA 4-digit - the tabulated profiles carry their own fixed camber.
Camber pos %	10 / 40 / 90	Position of maximum camber as % of chord from the leading edge. Limits 10-90 %. Typical 30-50 %. Ignored when camber is 0. Only shown for NACA 4-digit.
Root thick %	4 / 10 / 20	Root section thickness as % of chord. Limits 4-20 %. Typical 8-11 %. Thinner = less drag; below about 8 % stabs get flexible and stall abruptly.
Tip thick %	4 / 8 / 20	Tip section thickness as % of chord. Limits 4-20 %. Typical 7-10 %. Blend between root and tip is set by the thickness exponent.
TE thickness	0.1 / 0.6 / 3 mm	Limits 0.1-3 mm, and at most 3 % of the root chord. Thinned automatically near the tip. Typical 0.5-0.8 mm for wet-laid carbon in printed molds, 0.3-0.5 mm for prepreg. Thicker = tougher edge but more base drag and a risk of humming.

Label	Min / default / max	Meaning
Points/surface	10 / 36 / 120	Limits 10-120. Typical 30-40. Cosine-spaced towards the leading edge. Above about 60 slows regeneration for no visible gain.

Planform

Label	Min / default / max	Meaning
Planform input	dropdown: Area + aspect ratio, Span + root chord (default Area + aspect ratio)	Area + aspect ratio (recommended): enter the area everyone quotes and the shape; span and root chord are solved. Span + root chord: enter the geometry directly (e.g. to copy an existing foil); area becomes an output.
Area cm ²	30 / 165 / 1000	Target stab area in cm ² . Limits 30-1000 cm ² . Typical 15-22 % of the front wing: 150-350 cm ² for surf/eFoil/pump, 80-150 for kite race. Larger = more pitch stability and damping, more drag. The notice reports the ratio to the wing.
Aspect ratio	2 / 7.5 / 14	Limits 2-14, projected span ² / projected area. Typical 4-6 surf/eFoil, 6-9 pump/downwind/race. Higher = more efficient stab and more pitch damping per area, but more tip-breach risk and flex.
Area definition	dropdown: Projected area, Developed area (default Projected area)	Which area the target refers to. Projected (top view) is what sets lift and what most brands quote. Developed is measured along the curved span line and is larger with anhedral or gull. Both are reported.
Span	80 / 330 / 1200 mm	Limits 80-1200 mm, projected tip to tip (top view). Typical 250-450 mm.
Root chord	20 / 75 / 300 mm	Limits 20-300 mm. Typical 60-110 mm.
Chord distribution	dropdown: Superellipse, Linear taper (default Superellipse)	Superellipse: $c = c_0 (1 - (s \cdot 2y/b)^n)^{1/n}$, cut off at the end chord ratio, giving a rounded tip. Linear: straight taper to a square tip.
Exponent n	1.2 / 2.5 / 8	Limits 1.2-8. 2 = elliptic. 2.5-4 = fuller tips (common on stabs, keeps area outboard for damping). Above 5 approaches a rectangle with rounded corners. Typical 2.5-4.
End ratio	0.02 / 0.08 / 0.6	Limits 0.02-0.6. Chord of the small end face as a fraction of root chord. Typical 0.06-0.12 for a rounded tip (stab chords are small, so keep the end face above about 5 mm); 0.15-0.3 for a blunt tip.
Tip ratio	0.1 / 0.5 / 1	Limits 0.1-1.0. Tip chord of the straight taper as a fraction of root chord; the tip is square in plan view. Typical 0.4-0.7.
Min tip	checkbox, default off	On = the tip chord is widened automatically until the section at the tip cap start is at least Min tip mm thick, by raising End ratio (or Tip ratio) above the value entered. The notice reports the ratio used. Guards a thin, small-chord tip that cannot be laid up; especially useful on a stab linked to a wing, where the link sets area, aspect ratio and thickness but the tip chord is still this feature's own.
Min tip mm	0.2 / 1 / 5 mm	Minimum section thickness at the tip cap start (the last full section before the cap tapers out). Limits 0.2-5 mm. Typical 1-1.5 mm for a carbon layup: two skin plies plus a little UD; the cap beyond it is tow and resin. Only the tip chord is changed, never the thickness %.
Stations	3 / 11 / 30	Limits 3-30, per half span. Typical 9-13 for a rounded superellipse tip, 5-7 for linear taper. Tip-cap stations are added on top.
Mirror to full span	checkbox, default on	On = complete foil. Off = right half only, for faster edits and section inspection. Reported metrics are always full-span values.

Sweep, alignment and anhedral

Label	Min / default / max	Meaning
Align line %	0 / 50 / 100	Limits 0-100 %. Typical 25 (classic elliptical), 60-100 for modern pump/downwind and most stabs. The chord line that follows the sweep curve. 100 = straight trailing edge (very common on stabs), 25-50 = both edges curve, 0 = straight leading edge. Independent of the incidence pivot.
Sweep aft	-50 / 30 / 200 mm	Limits -50 to +200 mm, positive = aft. How far the alignment line sits aft at the tip. Typical 10-50 mm. More sweep = more yaw stability and a longer effective tail arm.
Sweep exp	0.5 / 2 / 4	Limits 0.5-4. Shape of the sweep line $x = \text{offset} (2y/b)^m$. 1 = straight (kinked at the root), 2-3 = curved and tangent at the root, 3-4 = raked tips. Typical 2.
Tip drop	-50 / 20 / 100 mm	Limits -50 to +100 mm, positive = down (anhedral). Typical 0-15 mm; negative (tips up) is used on some stabs for yaw stability. Large values can create mold undercuts.
Drop exp	1 / 2 / 6	Limits 1-6. Typical 2-3. Shape of the drop $z = -\text{drop} (2y/b)^e$. 1 = straight from the root, 2-3 = flat centre with tips falling away.
Gull dihedral	-10 / 0 / 20 deg	Limits -10 to +20 deg, positive = up. Gull term as on the wing; rarely used on stabs. Typical 0. Needs a drop exponent above 1.

Twist and thickness distribution

Label	Min / default / max	Meaning
Washout	-5 / 0 / 8 deg	Limits -5 to +8 deg, relative to the root incidence; positive = tip leading edge down. Typical 0 deg; stabs are normally untwisted.
Washout exp	0.5 / 1 / 4	Limits 0.5-4. Twist follows washout $(2y/b)^w$. 1 = linear, 2-3 = twist concentrated towards the tips. Typical 1.
Thickness exp	0.5 / 1.5 / 4	Limits 0.5-4. Thickness % follows root + (tip - root) $(2y/b)^p$. 1 = linear, 2-3 = keeps root thickness further out. Typical 1-2.

Incidence

Label	Min / default / max	Meaning
Incidence	-10 / -1.5 / 5 deg	Limits -10 to +5 deg, measured from the fuselage axis (X), positive = leading edge up. Typical -1 to -3 deg. Aim for a wing-minus-stab difference (decalage) of 1.5-3 deg; the notice reports geometric and aerodynamic decalage.
Pivot	dropdown: Quarter chord, Leading edge, Trailing edge (default Quarter chord)	Point each section rotates about for incidence and washout, and the line that follows the drop curve. The root pivot sits at the stab position. Quarter chord (recommended) keeps the lift centre fixed as angles change. Plan-view shape is set by the alignment line, not the pivot.

Position and link to wing

Label	Min / default / max	Meaning
Arm	200 / 650 / 1500 mm	Limits 200-1500 mm, positive = aft. Distance along the fuselage axis from the wing root pivot (world origin) to the stab root pivot. This sets the pitching moment arm and, with stab area, the tail volume - it is a design decision, not a fuselage detail. Typical 550-800 mm; short (450-550) for surf/carving, long (700-900+) for pump/downwind. Longer = more pitch stability and damping, slower pitch response. A fuselage, if you add one later, is built to match this arm rather than the other way round.

Label	Min / default / max	Meaning
Height	-100 / 0 / 100 mm	Where the stab's seat plane sits relative to the fuselage centreline (the wing's tube socket axis), positive = up; 0 = on the centreline. The seat plane is the pad face, or the bare root surface without a pad. Pad on top (fuselage over the stab): 0 cuts the fuselage flat to half the tube under the seat, -4 leaves a flat $2 \times \sqrt{R^2 - 16}$ wide and more tube (typical -3 to -5); below the axis only. Pad underneath (stab on top): +4 does the same the other way; above the axis only. The stab is placed to this after it is built, so resizing or relinking it never moves the seat. Without a tube socket on the wing this is the offset of the stab pivot from the wing pivot instead. Limits -100 to +100 mm. Typical -4 (pad on top), +4 (pad underneath), 0 to +20 with no socket.
Link to wing	dropdown: Independent, Linked to wing (default Independent)	Independent: every stab parameter below is typed in directly. Linked to wing: area, aspect ratio, thickness and incidence are derived from the wing's reported values instead, using the ratios and offsets below. Camber, planform shape, sweep, anhedral, twist and tip stay independent either way.
Area ratio	0.05 / 0.15 / 0.4	Limits 0.05-0.4. Stab projected area as a fraction of the wing's projected area. Typical 0.15-0.22. Replaces the direct area input; aspect ratio still sets the span/chord split.
AR offset	-8 / -1.5 / 4	Limits -8 to +4. Added to the wing's aspect ratio to get the stab's. Typical -2 to 0: a lower-AR stab is common for extra pitch damping and robustness.
Thick offset	-6 / -2 / 2	Limits -6 to +2 %. Added to the wing's root and tip thickness % to get the stab's, so the stab reliably stays thinner (or thicker) than the wing. Typical -3 to -1.
Decalage	0 / 1.5 / 6 deg	Limits 0-6 deg. Stab incidence is set to the wing's incidence minus this value, so you tune the one number that actually sets trim instead of two absolute angles. Typical 1.5-3 deg geometric. The notice still reports the resulting geometric and aerodynamic decalage.

Tip

Label	Min / default / max	Meaning
Tip cap	dropdown: Rounded tip, Flat tip (default Rounded tip)	Rounded tip: section thickness falls along a quarter-ellipse over the cap length, rounding the tip in front view. Plan-view rounding comes from the superellipse end ratio. Flat tip: thickness is unchanged to the end face.
Cap length	2 / 8 / 40 mm	Limits 2-40 mm, and at most 30 % of the half span. Typical 5-12 mm.
Cap end %	5 / 30 / 80	End-face thickness as % of the tip section thickness. Limits 5-80 %. Typical 20-35 % (stab tips are small; keep the end face layable).

Section polar

Label	Min / default / max	Meaning
Section data	dropdown: Analytic, Polar summary, Built-in (default Analytic)	Built-in: fitted XFOil data, nothing to type in. E817/E818/NACA 63-412 use one real polar each. NACA 4-digit uses the exact zero-lift angle/moment for your Camber %/Camber pos %, plus lift slope, CLmax and drag bucket fitted from real NACA polars at your thickness/camber - coarser than the fixed profiles (a fit across a family, not one measured shape); use Polar summary for higher accuracy at your exact section. Analytic (default): thin-aerofoil model - camber-line angle/moment, fixed lift slope, skin-friction drag, Section CL max from Foil performance. Polar summary: your own XFOil/XFLR5 numbers at your own Reynolds numbers, via the fields below. Run scripts/polar_summary.py (developer skill) on your polar - the tool never invents section data.

Label	Min / default / max	Meaning
Zero-lift	-8 / -2 / 4 deg	Zero-lift angle of attack from the polar, deg (usually negative for a cambered section). Limits -8 to 4 deg. Replaces the thin-aerofoil estimate. Typical -1 to -3 deg for a few % camber. For the built-in profiles (E817, E818, NACA 63-412) use Section data = Built-in instead of typing these in.
Lift slope	4 / 6 / 7.5	Section lift-curve slope from the polar, per radian. Limits 4-7.5. Thin-aerofoil theory gives $2\pi = 6.28$. Replaces the fixed analytic value. Typical 5.5-6.5 depending on thickness and Reynolds number. For the built-in profiles (E817, E818, NACA 63-412) use Section data = Built-in instead of typing these in.
Cm c/4	-0.15 / -0.05 / 0.05	Section pitching moment about the quarter chord from the polar (usually fairly constant with CL below stall). Limits -0.15 to 0.05. Replaces the thin-aerofoil Cm0. Typical -0.03 to -0.11 for a cambered section, near 0 for symmetric. For the built-in profiles (E817, E818, NACA 63-412) use Section data = Built-in instead of typing these in.
Re 1 (M)	0.1 / 0.2 / 5	First Reynolds number the polar data below is from, in millions. Limits 0.1-5. Values are interpolated (log Re) between Re 1 and Re 2, and held flat outside that range. Typical: near your take-off Reynolds number (about 0.2-0.5 M for a wing, lower for a stab). For the built-in profiles (E817, E818, NACA 63-412) use Section data = Built-in instead of typing these in.
CLmax @Re1	0.6 / 0.85 / 1.8	Maximum section lift coefficient at Re 1. Limits 0.6-1.8. Real CLmax usually drops noticeably below the higher-Re value, which is why a single fixed CL max (the analytic fallback) can be optimistic at take-off. Typical 0.85-1.1 at low Re. For the built-in profiles (E817, E818, NACA 63-412) use Section data = Built-in instead of typing these in.
Cdmin @Re1	0.004 / 0.01 / 0.03	Minimum section drag coefficient at Re 1 (the bottom of the drag bucket). Limits 0.004-0.03. Friction drag rises as Re falls. Typical 0.008-0.018 at low Re. For the built-in profiles (E817, E818, NACA 63-412) use Section data = Built-in instead of typing these in.
CL@Cdmin 1	-0.3 / 0.3 / 1	The lift coefficient at which Cdmin occurs at Re 1 - the centre of the drag bucket. Limits -0.3 to 1.0. Typical 0.3-0.7. For the built-in profiles (E817, E818, NACA 63-412) use Section data = Built-in instead of typing these in.
k @Re1	0.005 / 0.013 / 0.05	Drag-rise rate at Re 1: $C_d = C_{dmin} + k (C_L - C_{L@Cdmin})^2$. Limits 0.005-0.05. For your own polar, read it off two points on the drag bucket away from Cdmin and solve for k, or use polar_summary.py to fit it automatically. Typical 0.005-0.04. For the built-in profiles (E817, E818, NACA 63-412) use Section data = Built-in instead of typing these in.
Re 2 (M)	0.1 / 0.5 / 5	Second Reynolds number, in millions. Limits 0.1-5. Must differ from Re 1 for interpolation to do anything. Typical: near your cruise or max-speed Reynolds number. For the built-in profiles (E817, E818, NACA 63-412) use Section data = Built-in instead of typing these in.
CLmax @Re2	0.6 / 1 / 1.8	Maximum section lift coefficient at Re 2. Limits 0.6-1.8. Typical 1.0-1.3 at higher Re. For the built-in profiles (E817, E818, NACA 63-412) use Section data = Built-in instead of typing these in.
Cdmin @Re2	0.004 / 0.007 / 0.03	Minimum section drag coefficient at Re 2. Limits 0.004-0.03. Typical 0.004-0.008 at higher Re. For the built-in profiles (E817, E818, NACA 63-412) use Section data = Built-in instead of typing these in.
CL@Cdmin 2	-0.3 / 0.3 / 1	The lift coefficient at which Cdmin occurs at Re 2. Limits -0.3 to 1.0. Typically close to CL@Cdmin 1, since the drag bucket centre moves less with Re than CLmax does. Typical 0.3-0.4. For the built-in profiles (E817, E818, NACA 63-412) use Section data = Built-in instead of typing these in.
k @Re2	0.005 / 0.01 / 0.05	Drag-rise rate at Re 2. Limits 0.005-0.05. Typical 0.01-0.035. For the built-in profiles (E817, E818, NACA 63-412) use Section data = Built-in instead of typing these in.

Label	Min / default / max	Meaning
Invert section	checkbox, default on	Mount the stab's tabulated profile (E817, E818, NACA 63-412) upside down, camber down, so it makes the downforce a stab is there for - the convention for a cambered stab section. The right way up it LIFTS at cruise and throws the trim CG far aft. No effect on NACA 4-digit, where the sign of Camber % does this (negative = inverted).

Mount pad

Label	Min / default / max	Meaning
Mount pad	checkbox, default on	Adds the production-style mount to the stab root, modelled on the Axis rear wing: a flat pad face on the fuselage side, faired in from the leading edge, the full root chord long, with an optional tail fairing running aft of the trailing edge to close the fuselage's end. The Foil fuselage seats on this face and the Foil mold puts bolt cones in it. Off = bare root, the fuselage seats on the curved skin.
Pad side	dropdown: Top (fuselage above), Bottom (fuselage below) (default Top (fuselage above))	Top: the fuselage passes over the stab and bolts up through it into the fuselage (Axis, Lift, Armstrong layout). Bottom: the stab sits on top of the fuselage tail and bolts down into it. Top is the norm: the bolt heads sit in the stab's underside, the fuselage tail fairs into the stab's tail, and the stab's convex side faces down.
Pad width	5 / 22 / 80 mm	Width of the pad face across the span, centred on the root. Limits 5-80 mm. Typical 20-24 mm (Axis 22); match or slightly under the fuselage Tail dia so the seat is full width. Also the width of the tail fairing where it leaves the trailing edge.
Pad height	1 / 6 / 20 mm	How far the flat pad face stands above the section's highest point at the root (on the pad side). Limits 1-20 mm. Typical 5-8 mm (Axis 7.4): room for the fuselage seat to be full width and the bolt heads or threads to bite. Sets the pad plane the fuselage seats on.
Flat start	-100 / -5 / 100 mm	Where the flat pad face begins, in mm from the stab pivot along X (negative = ahead of it). Limits -100 to 100 mm. Typical -10 to 0 (just ahead of the quarter-chord pivot). The ramp fairs up from the leading edge to here, so it adapts to the stab; everything aft of here is the fixed interface the fuselage seats on. Must be at least 5 mm behind the leading edge - the notice warns and holds it there if not.
Tail fairing	checkbox, default on	Continues the pad aft of the trailing edge as a tail: full Pad width from the trailing edge to Tail start, then tapering to End width x End thick at Tail end, flat on the pad side. Its top is the same seat plane, so the fuselage's flat runs from Flat start to Tail end and its tip tapers from Tail start, whatever the stab's chord. Off = the pad ends at the trailing edge and the interface is no longer fixed.
Tail start	0 / 85 / 300 mm	Where the tail fairing and the fuselage tip begin to taper, in mm aft of the stab pivot. Limits 0-300 mm. Typical 80-100: at or a little behind the trailing edge of your biggest stab (root chord x 0.75 from a quarter-chord pivot). A stab whose trailing edge is behind this is warned and the taper starts at its trailing edge instead.
Tail end	10 / 125 / 400 mm	Where the tail fairing and the fuselage end, in mm aft of the stab pivot. Limits 10-400 mm. Typical Tail start + 35-50 (Axis tail 40 mm). Must be at least 5 mm behind Tail start and the trailing edge.
End width	1 / 4 / 30 mm	Plan width of the tail at its end. Limits 1-30 mm. Typical 3-6 mm (Axis 4). The taper from Pad width is concave - fast at first, then easing - like a fuselage tail.
End thick	1 / 6 / 20 mm	Thickness of the tail at its end, from the pad plane to the far surface. Limits 1-20 mm. Typical 5-7 mm (Axis 6): the far surface slopes from the trailing edge up to here.

Mount bolts

Label	Min / default / max	Meaning
Stab bolts	0 / 2 / 6	Through holes on the centreline for the bolts that hold the stab to the fuselage. Limits 0-6. Typical 2 (Axis: two at 35 mm). The stab owns this pattern: the Foil fuselage taps its blind holes to match and the Foil mold puts undersized cones at the same positions. 0 = no holes.
Bolt 1 at	-100 / 0 / 200 mm	First bolt's position in mm from the stab pivot along X (negative = ahead). Limits -100 to 200 mm. Typical 0 (Axis: 22 mm behind the LE on an 89 mm chord, i.e. at the quarter chord, where the section is thickest). Fixed to the pivot, so the fuselage's tapped holes are the same for every stab. Keep it on the flat (at or behind Flat start) for a seat of constant depth.
Bolt pitch	5 / 35 / 200 mm	Spacing between bolts along X. Limits 5-200 mm. Typical 30-40 mm (Axis 35). Every bolt must sit inside the root or the tail; the notice warns if one does not.
Bolt	dropdown: M4, M5, M6, M8, M10, Custom hole sizes (default M6)	The bolt that holds the stab to the fuselage. Sets the stab's clearance hole (close fit: M4 4.4, M5 5.4, M6 6.4, M8 8.4, M10 10.5 mm) and, through the published pattern, the Foil fuselage's tapping drill (3.3 / 4.2 / 5.0 / 6.8 / 8.5 mm), so the two can never disagree. Typical M6 (Axis), M8 for a big stab. Custom hole sizes: type the clearance hole here and the tapping drill in the fuselage yourself.
Bolt dia	2 / 8.4 / 14 mm	Clearance hole diameter in the stab (Custom only). Limits 2-14 mm. Typical 8.4 (M8) or 6.4 (M6). This is the finished hole: the mold makes an undersized cone and it is drilled to this after cure. Set the matching tapping drill in Foil fuselage's Stab bolt dia.
Countersink	checkbox, default on	90 deg countersink on the face away from the pad, where the bolt heads sit (the underside with the pad on top), to the ISO 10642 head diameter of the chosen Bolt (M6 13.4, M8 17.9 mm; Custom: twice the hole). Cut in the model for fit; on the molded part it is drilled and countersunk after cure like the hole itself. Skipped for a bolt behind the trailing edge.
Log inputs	checkbox, default on	Prints every parameter on this feature to the FeatureScript console as one line, so the whole set can be pasted elsewhere (e.g. back to an AI assistant) along with the notice or CSV.

Foil performance

Load and water

Label	Min / default / max	Meaning
Total mass kg	20 / 75 / 250	Rider + board + foil (+ battery for eFoil), in kg. Limits 20-250. Typical 75-110 kg for an adult rider with board and foil; add 10-20 kg of battery for eFoil. The weight the foil must lift at every speed.
Pitch Nm	-500 / 0 / 500	External pitching moment about the centre of mass, N m, positive = nose up. Limits -500 to +500. Typical 0 unless modelling propulsion or sail forces. 0 for pump/SUP/surf (rider weight only). eFoil: with Propulsion on and a motor pod, the thrust/drag couple is added automatically, so leave this 0 unless something else acts. Wind/kite: sail or harness force times its height above the CG. Shifts the trim CG position; does not change drag.
Density kg/m ³	990 / 1025 / 1040	Water density. Limits 990-1040. Typical 1025 (sea) or 998 (fresh). 1025 seawater, 998 fresh water. Fresh water needs about 1.3 % more speed for the same lift.
Water temp C	0 / 20 / 35	Water temperature in deg C. Limits 0-35. Typical 15-25 C. Sets viscosity (Reynolds number, skin friction) and vapour pressure (cavitation). Cold water raises friction drag slightly.

Label	Min / default / max	Meaning
Wing depth	100 / 600 / 3000 mm	Depth of the front wing below the surface. Limits 100-3000 mm. Typical 400-800 mm when riding. Used for the cavitation number and a free-surface warning: below about 2 MAC the surface causes lift loss and wave drag that are not modelled.

Speeds

Label	Min / default / max	Meaning
Take-off km/h	3 / 15 / 60	Speed at which you want to be foiling. Limits 3-60 km/h. Typical 8-12 pump/SUP, 12-16 eFoil/surf, 12-18 wind, 15-25 kite race. The notice shows how close to stall the wing is here.
Cruise km/h	4 / 20 / 80	Main operating speed; headline L/D, drag, power, trim and structure are reported here. Limits 4-80 km/h. Typical 12-18 pump/downwind, 18-25 eFoil/wind, 30-45 kite race.
Max km/h	5 / 30 / 100	Top of the analysed range, used for the cavitation check. Limits 5-100 km/h. Typical 25-35 eFoil/wind, 45-70 kite race.
Step km/h	0.5 / 1 / 5	Speed step for the CSV sweep printed to the FeatureScript console, from take-off to max speed. Limits 0.5-5 km/h. Typical 1 km/h.

Section and drag model

Label	Min / default / max	Meaning
Section CL max	0.6 / 1 / 1.8	Maximum lift coefficient of the section. Used only when the wing's Section polar (in Foil wing) is set to Analytic - if it is set to Polar summary, the polar's own CLmax (interpolated at the take-off Reynolds number) is used instead and this field is ignored. Limits 0.6-1.8. Typical 1.0-1.2 for 8-10 % thick low-camber sections, 1.2-1.4 for 12 % with 2-3 % camber, 1.4-1.6 for 14 % with 4 % camber. Real values drop at low Reynolds number. Local lift in the outer 5 % of span is ignored.
Laminar run %	0 / 0 / 60	Portion of the chord in laminar flow before transition. Limits 0-60 %. Typical 0-20 % for molded carbon parts. 0 = fully turbulent (conservative; realistic for a hand-finished or scratched surface). 20-40 % for a smooth, fair finish on a NACA 4-digit; 50-60 % only for laminar sections with a polished finish.
Other cm ²	0 / 0 / 200	Extra drag area (Cd x area) in cm ² for anything NOT modelled. Limits 0-200. Typical 1-4 cm ² . The mast (submerged part + spray), motor pod and fuselage are now computed from the Foil mast and Foil fuselage features, so do not include them here; use this for junctions, the wing socket nacelle, fittings and bolt heads.

Structure

Label	Min / default / max	Meaning
Load factor g	1 / 3 / 8	Multiple of the cruise wing lift used for the root bending check. Limits 1-8. Typical 3 for general riding, 4-5 for pumping, jumps and hard landings, 6+ for kite freestyle.
Cap stress MPa	100 / 450 / 1500	Allowable stress in the unidirectional carbon spar caps, MPa. Limits 100-1500. Typical 300-500 for wet-laid UD carbon with a safety factor (compression governs, roughly half the tensile strength), 600-900 for prepreg. Lower = more conservative.

Label	Min / default / max	Meaning
Cap width %	10 / 35 / 70	Width of each spar cap as % of root chord. Limits 10-70 %. Typical 25-40 %. Wider caps can be thinner for the same strength.

Propulsion

Label	Min / default / max	Meaning
Propulsion	checkbox, default off	eFoil / foil-assist drivetrain. At steady speed prop thrust equals foil drag, so from the drag at each speed this finds the prop rpm (Wageningen B-series open-water KT/KQ, the same polynomials as the B-series blade feature), then motor current, throttle, battery power, Wh/km, range, top speed, and the thrust/drag couple in trim (thrust on the pod axis). Off for pump, SUP, surf, wing, wind and kite. Open-water estimate, about +/- 20 %: no pod or mast wake, no free surface, no cavitation or ventilation.
Prop dia	60 / 140 / 400 mm	Propeller diameter. Limits 60-400 mm. Typical 120-160 mm for eFoil, 100-140 mm for foil assist. The prop axis is taken as the motor pod axis for the immersion check. Bigger turns slower and more efficiently but needs more depth to stay submerged.
Prop P/D	0.5 / 0.8 / 1.4	Pitch / diameter at 0.7R. Limits 0.5-1.4 (the B-series fitted range). Typical 0.8-1.0 for eFoil. Higher pitch raises top speed and cruise efficiency, lower pitch gives more static thrust. Top speed is roughly pitch x no-load rpm less slip.
Prop EAR	0.3 / 0.3 / 1.05	Expanded blade area ratio. Limits 0.3-1.05 (the B-series fitted range). Typical 0.3-0.55 for eFoil. More area spreads the load (less cavitation) but adds blade friction.
Blades	2 / 3 / 7	Number of blades. Limits 2-7 (the B-series fitted range). Typical 2-3 for eFoil.
Motor kV	50 / 450 / 1000	Motor speed constant, rpm per volt at no load. Limits 50-1000. Typical 150-500 for eFoil outrunners, geared or direct. Torque per amp is taken as 8.27 / kV N m (same as the blade feature).
Cells (S)	1 / 12 / 30	Lithium cells in series; pack voltage = cells x 3.7 V nominal. Limits 1-30. Typical 10-14 S for eFoil (12 S = 44.4 V).
Gear ratio	1 / 4 / 10	Motor turns per prop turn. Limits 1-10. Typical 1 (direct drive) or 3-5 for a geared outrunner. Higher gives more prop torque, less prop rpm.
Current lim A	1 / 120 / 600	Motor current limit set in the ESC, A. Limits 1-600. Typical 80-150 A for eFoil. Sets the full-power torque, so the static thrust and the top speed.
Winding mOhm	0 / 12 / 500	Motor winding (phase) resistance, milliohm. Limits 0-500. Typical 5-40 for eFoil motors; 12 matches the B-series blade feature's check of a 450 kV 12 S drive. Near no-load rpm it caps the current the pack voltage can push. 0 = ignore (optimistic top speed).
Gearbox eff %	80 / 95 / 100	Gearbox efficiency, %. Limits 80-100. Typical 95 for one planetary stage, 100 for direct drive. The blade feature ignores gear loss, so set 100 to compare with its numbers.
ESC eff %	80 / 97 / 100	ESC efficiency, %. Limits 80-100. Typical 95-98. Motor losses are already in the 8.27 / kV torque constant and the winding resistance.
Pack Ah	1 / 20 / 100	Battery pack capacity, amp hours. Limits 1-100. Typical 12-30 Ah for eFoil. With Cells (S) x 3.7 V and Usable % this sets the range.
Usable %	50 / 90 / 100	Share of the nameplate pack energy you actually use before cut-off or voltage sag ends the ride, %. Limits 50-100. Typical 80-90.
Print report	checkbox, default on	Prints the full report and a CSV speed sweep to the FeatureScript console, ready to paste into a spreadsheet.

Foil mold

Part

Label	Min / default / max	Meaning
Part	selection	The wing or stabiliser solid to mold (the whole body, tube socket or mount pad included). One Foil mold per component - add a separate instance for the wing and for the stab.
Source model	dropdown: Wing, Stabiliser, Mast (default Wing)	Which feature built Part; it must match the body selected above. Wing / Stabiliser: the parting follows that feature's own planform, twist, droop and section. Mast: the mold is built on a copy of the mast laid flat 400 mm below the origin (span along -Y, thickness up), parting on the mast's mid-plane, in three parts (left, right and a tray for the plate); the plate is molded with it (drill its holes after cure) and the motor pod is molded too, with a plug shaped from the pod's cavity.
Parts group	dropdown: Closed: one entry in the parts list, Open: parts stay listed, group under Composite parts, Off: loose parts (default OFF)	Off (default): every mold piece is a normal part, named. Closed / Open collect everything this feature makes - shells, tiles, plugs, pins and the template plate - into one composite part named after the source (Wing mold, Stab mold, Mast mold). Closed: the parts list shows one expandable entry and the pieces live inside it (expand it to reach a tile; export from there). Open: the pieces stay listed as parts and the group appears under Composite parts, which is how Onshape treats open composites. Closed can hide every piece inside a group that is itself hidden, which is why the default is Off.
Split stations	5 / 15 / 80	Spanwise stations used to build the parting surface, root to tip. Limits 5-80. Typical 15-25; 40-80 for a tip with all its sweep and droop concentrated in the last fifth of the span (Sweep exp / Drop exp of 3-4), where the parting must turn sharply. Each station is a sketch and a ruled loft, so regeneration time grows with the count. Packed toward the tip by Tip bias.
Tip bias	1 / 2 / 4	How strongly the stations crowd toward the tip: station i of n sits at span fraction $1 - (1 - i/n)^{\text{bias}}$. Limits 1-4. Typical 2. 1 = evenly spaced; 2 = close to the cosine packing used before; 3-4 = most stations in the outer third, for planforms whose sweep and droop are concentrated there. Raise this before raising Split stations.

Mold body

Label	Min / default / max	Meaning
Flange width	10 / 25 / 60 mm	Mold material around the planform, from the part's edge to the outside of the block. Limits 10-60 mm. Typical 20-30 mm for a clamped printed mold: wide enough to seat clamps and a pinch-off groove, narrow enough not to waste print time. The parting surface runs out through the flange, so the two halves meet on it all the way round.
Base thickness	5 / 20 / 40 mm	Mold material between the cavity and the outer (clamped) face of each half, measured at the deepest point. Limits 5-40 mm. Typical 15-25 mm for a printed mold clamped against a rigid board: enough not to flex between clamps or print through to the cavity, without a long print, and it is the layer the tile keys sit in (at least Pin dia + 4 mm and Key height + 4 mm). The slicer sets walls and infill; the body is solid here.

Tube socket

Label	Min / default / max	Meaning
Socket plug	checkbox, default on	Wing molds only, when the Foil wing has its Tube socket on. The socket's bore is not molded by the halves (two half-round cores would leave a seam down the bore): the cavity is cut with the bore and pin holes filled, and a separate cylindrical plug part - the bore diameter, the socket's full depth plus Plug extension - is made to lay the carbon around. Both halves get a half-round channel behind the nacelle's rear face that holds the plug coaxial when the mold is closed; pull the plug after cure and drill the pin holes to the wing's pattern. Off = the bore and pin holes are molded as half cores (not recommended).
Plug extension	10 / 20 / 60 mm	How far the plug reaches behind the nacelle's rear face into the channel in the flanges. Limits 10-60 mm, and at most Flange width - 3 mm. Typical 15-20 mm: enough to hold it square and to grip when pulling. Pin fit sets the channel clearance.

Bolt cones

Label	Min / default / max	Meaning
Bolt cones	checkbox, default on	Stabiliser molds only. Puts a tapered cone at each of the stab's own Mount bolts, standing on the pad face in the half that holds the pad and pointing through the section to Cone web short of the far skin, so the holes come out of the mold on the pattern the Foil fuselage is tapped to. Undersized and tapered so the part releases; drill to the stab's Bolt dia after cure. Off = drill from the fuselage as a jig.
Cone dia	2 / 5 / 12 mm	Diameter of each cone at the pad face, the wide end. Limits 2-12 mm. Typical 4-5 mm for M6-M8, leaving 1.5-2 mm a side to drill out to clearance. Keep it below the fuselage's Stab bolt dia so the drilled hole cleans the molded surface.
Cone taper	2 / 6 / 15 deg	Half-angle of the cone. Limits 2-15 deg. Typical 5-8 deg: enough draft to pull a clamped part off without levering, on a hole that will be drilled anyway.
Cone web	0.5 / 1 / 5 mm	Laminate left between the cone tip and the top surface, so the cone never touches the upper half and the hole is closed by a thin web you drill through. Limits 0.5-5 mm. Typical 1-1.5 mm.

Pinch-off

Label	Min / default / max	Meaning
Pinch-off groove	checkbox, default on	A channel in the lower half's parting face running right round the cavity: a narrow land at the cavity edge where the closing mold pinches the laminate and squeezes the excess resin and fibre out, then a groove for that flash to run into. Follows the planform (leading edge, tips, trailing edge, the tube socket nacelle) and the parting surface down the dropped tips. Off for a mold you will bag rather than clamp.
Land width	1 / 3 / 10 mm	Flat land between the cavity edge and the groove. Limits 1-10 mm. Typical 2-4 mm: narrow enough to build real pinch pressure, wide enough to print cleanly and survive the clamps.
Groove width	1 / 4 / 10 mm	Width of the flash groove. Limits 1-10 mm. Typical 3-5 mm.
Groove depth	0.5 / 1.5 / 5 mm	Depth of the flash groove below the parting face. Limits 0.5-5 mm. Typical 1-2 mm - enough volume for the squeeze-out of an 8 % overcharge, shallow enough to leave the flange stiff.

Printer

Label	Min / default / max	Meaning
Printer X	50 / 250 / 1000 mm	Build volume across the chord (tile width). Limits 50-1000 mm. Typical 200-300. The report shows how many tiles each half needs; a tile can be rotated on the bed, so enter the bed's shorter side here and its longer side as Printer Y.
Printer Y	50 / 250 / 1000 mm	Build volume along the span (tile length). Limits 50-1000 mm. Typical 200-300.
Printer Z	50 / 250 / 1000 mm	Build height. Limits 50-1000 mm. Typical 200-300. Each half is printed cavity-up, so this only has to clear the half's own height (cavity depth plus Base thickness).
Tile margin	0 / 5 / 30 mm	Clearance kept between a tile and the printer's limits in every axis. Limits 0-30 mm. Typical 5 mm for brims, skirts and bed-edge adhesion.
Stagger joints	checkbox, default on	Offsets the upper half's spanwise cuts by half a tile from the lower half's, so no joint runs through both halves at the same place: the assembled mold braces itself and the seams never line up through the laminate. The upper half then has one extra tile (two half-length end tiles). Off = the same cuts in both halves.

Joints

Label	Min / default / max	Meaning
Key style	dropdown: Printed keys, Dowel holes (default Printed keys)	How tiles locate on each other and the halves on one another. Printed keys: a tapered round boss on one side of every joint and a matching socket on the other, printed with the mold - nothing to buy. Bosses always point +Y (span joints), +X (chord joints) and up from the lower half, so they print without support with the halves cavity-up. Dowel holes: a plain hole either side of every joint for a steel or wooden dowel.
Pin dia	3 / 12 / 30 mm	Diameter at the base of each tile-joint key (or the tile dowel). Limits 3-30 mm, and at most Base thickness - 4 mm (the keys sit in the base layer) and Flange width - 4 mm. Typical 10-16 mm for printed keys, 6-8 mm for steel dowels. Two per spanwise joint sit in the front and rear flanges; one per tile crosses each chordwise joint under the cavity. All of them sit in the base layer of each half, so none can break into the cavity.
Half keys	checkbox, default on	Registration between the two halves: a key (or dowel hole) through the parting surface in the front and rear flanges, one pair per spanwise tile a quarter tile in from the tile joint - clear of every cut and tile key in both halves. Printed keys stand up from the lower half into sockets in the upper. Off = the halves locate on the clamps and board alone.
Key dia	3 / 14 / 30 mm	Base diameter of the half-to-half keys (or dowel). Limits 3-30 mm, and at most Flange width - 4 mm. Typical 12-20 mm for printed keys, 6-8 mm for dowels.
Key height	2 / 8 / 25 mm	Height of every printed boss above its face; the socket is 0.3 mm deeper so the boss bottoms on the joint face, not in the socket. Limits 2-25 mm, and at most Base thickness - 4 mm (the socket must not break through the other side). Typical 8-12 mm. Taller locates better but the tip must stay over 2 mm across (see Key draft).
Key draft	3 / 12 / 30 deg	Taper of the boss sides, so it finds its socket as the parts come together. Limits 3-30 deg. Typical 10-15 deg. Tip diameter = base - 2 x height x tan(draft), and must stay over 2 mm.
Pin length	10 / 24 / 60 mm	Total tile dowel length, half in each tile. Limits 10-60 mm. Typical 20-30 mm.
Key length	8 / 16 / 40 mm	Total half-to-half dowel length, half in each half. Limits 8-40 mm, and at most 2 x (Base thickness - 3 mm) so the hole never breaks through the thinner half at a dropped tip. Typical 16-20 mm.

Label	Min / default / max	Meaning
Pin fit	0 / 0.2 / 0.5 mm	Clearance added on the radius of every socket over its key, or to the hole diameter over the dowel. Limits 0-0.5 mm. Typical 0.15-0.25 mm for printed key in printed socket (add more if your printer prints holes small); 0.1-0.2 for a steel dowel.

Layup

Label	Min / default / max	Meaning
Skin plies	0 / 2 / 10	Bidirectional (woven) plies on each face of the part, full planform, laid first against the mold. Limits 0-10. Typical 1-2 for a foil wing or stab, 2-3 for a mast. They give the surface finish and torsional stiffness; the UD caps and tow carry the bending. Sized from the part's own surface area.
Skin gsm	50 / 200 / 800	Areal weight of the woven skin cloth, g/m2. Limits 50-800. Typical 160-245 (2x2 twill), 100-200 for a fine finish. Cured ply thickness is derived from this and Fibre vol %: $\text{gsm} / (1780 \times \text{Vf})$, which gives 0.22 mm for 200 gsm at 50 %.
UD gsm	50 / 300 / 1200	Areal weight of the unidirectional spar-cap fabric, g/m2. Limits 50-1200. Typical 200-300 wet layup, 150-300 prepreg, 400-600 for a thick root. Sets the cured UD ply thickness ($\text{gsm} / (1780 \times \text{Vf})$) and so the cap ply count.
Cap width %	10 / 40 / 70	Width of the UD spar caps as % of local chord, centred on the section's thickest point. Limits 10-70 %. Typical 35-50 %. Wider caps move fibre from tow into UD, which is where bending strength comes from, at the cost of more UD to cut.
Cap thick %	0 / 60 / 100	Thickness of the two UD caps together (top + bottom) as % of the core thickness at the thickest point - the local section less both skins. Limits 0-100 %. Typical 50-70 %. The cap ply count follows the section taper, dropping plies along the span - the console prints the cut list, per cap: cut every UD template twice, one for each face. What the skins and caps do not fill is filled with tow, so fewer skin plies means more UD at the same %.
Tow tex	100 / 800 / 4000	Linear density of the carbon tow (roving) that fills the lens core, leading and trailing edges and the socket nacelle, g/km. Limits 100-4000. Typical 800 (12k); 3k = 200, 24k = 1600 tex. Used only to turn the tow mass into a length to buy.
Fibre vol %	30 / 50 / 65	Fibre volume fraction of the cured laminate. Limits 30-65 %. Typical 45-50 wet layup clamped in a matched mold, 55-60 vacuum-bagged or prepreg, 35-40 wet hand layup with no pressure. Sets the cured ply thicknesses and the split of the part's volume between carbon (1.78 g/cm3) and epoxy (1.15 g/cm3).
Overcharge %	0 / 8 / 30	Extra material laid over the exact cavity fill, as %. Limits 0-30 %. Typical 5-10 % for a clamped matched-die mold: closing under pressure squeezes the excess out as flash at the pinch-off, which is what packs the laminate void-free. 0 for a loose vacuum bag. Scales the ply counts and every material figure.
Cloth waste %	0 / 25 / 100	Cutting waste added to the woven and UD areas to buy. Limits 0-100 %. Typical 20-30 % (bias cuts, tapered UD plies, selvage).
Resin waste %	0 / 40 / 150	Resin mixed beyond what stays in the part: mixing-pot loss, wet-out of the tow, brushes, and the flash squeezed out at the pinch-off. Limits 0-150 %. Typical 30-50 % wet layup, 10 % prepreg (only the tow needs wetting).
Cap run-out	0 / 0 / 100 mm	Mast only. Length each UD cap ply runs past the plate's underside, folded outward over the root fillet onto the plate between its fill plies, so the cap's tension enters the plate by shear instead of peeling off the corner. Limits 0-100 mm. 0 = as far as the plate edge allows (2 mm short of it). Typical 30-50 mm; the stiffness line says how much the junction needs.

Label	Min / default / max	Meaning
Tab plies	0 / 4 / 12	Mast only. UD tab plies per face interleaved with the cap plies at the plate: each runs Tab length down the mast face and Cap run-out out over the plate, alternating with the caps so the two surfaces are stitched together through the stack rather than by one bond line. Limits 0-12. Typical 3-5. They take their thickness from the tow core over Tab length; the layup line warns if that core runs out.
Tab length	10 / 50 / 200 mm	Mast only. How far each tab ply runs down the mast face from the plate. Limits 10-200 mm. Typical 40-80 mm, at least the Cap run-out.

Stiffness

Label	Min / default / max	Meaning
UD GPa	80 / 120 / 200	Tensile modulus of the cured UD carbon along the fibre. Limits 80-200. Typical 110-130 at 50 % fibre volume wet layup, 140-160 prepreg, 200+ for high-modulus fibre. Sets the spar caps' contribution to bending stiffness.
Woven GPa	30 / 60 / 100	In-plane modulus of the cured woven skin along the span. Limits 30-100. Typical 55-65 for a 0/90 twill at 50 % fibre volume; about 15 if the skin is laid at 45 deg, in which case it adds torsion, not bending.
Tow GPa	10 / 30 / 80	Effective modulus of the tow-filled core along the span. Limits 10-80. Typical 30 for tow laid roughly along the span, 10 for chopped or random fill.
Mast side g	0.2 / 1 / 3	Mast load case: a sideways force of this many times the all-up weight, applied at the fuselage end with the plate held by the board and rider, bending the mast about its chord - a hard carve or a fall. Limits 0.2-3. Typical 0.5-1. Reported as the deflection at the fuselage end and the UD cap stress at the plate, with the mast's own layup, taper and thickening; the moment peaks at the plate, which is why the top is thickened.
Shear allow	10 / 30 / 80	Interlaminar shear strength of the cured laminate, MPa, for the junction check at the mast plate: the cap force at the plate divided by the bonded run-out area (the cap's plate face plus both faces of every tab ply) against this value with a factor of 2. Limits 10-80. Typical 25-35 wet layup, 50-70 prepreg. The line says the run-out the junction needs at this allowable.

Cut templates

Label	Min / default / max	Meaning
Cut templates	checkbox, default off	Makes a flat 1 mm plate beside the mold with every fabric shape cut through it: the woven skin outline (the planform developed along the droop, cut once per skin ply per face) and one tapered strip per UD cap ply from the cut list (cut twice, one per face - the tally holes say so), each grown by Trim margin. Right-click the plate's top face in Onshape and Export as DXF to get every outline in one file for a cutter or a printed paper template. The console prints the order they appear in. Off = no plate.
Trim margin	0 / 5 / 20 mm	Allowance added round every template outline, for trimming and for the fabric's own slack. Limits 0-20 mm. Typical 5 mm woven, 3-5 mm UD.
Template gap	5 / 10 / 40 mm	Space between neighbouring outlines on the plate. Limits 5-40 mm. Typical 10 mm.
Log inputs	checkbox, default on	Prints every parameter on this feature to the FeatureScript console as one line, so the whole set can be pasted elsewhere (e.g. back to an AI assistant) along with the notice.

Foil mast

Mast

Label	Min / default / max	Meaning
Height	300 / 750 / 1300 mm	Mast length from its base (top of the fuselage) to the underside of the board plate. Limits 300-1300 mm. Typical 600-1000 mm; eFoil masts often 650-850 mm. Longer = more clearance in chop, more drag and more bending at the base.
Width	60 / 120 / 200 mm	Mast chord (fore-aft width). Limits 60-200 mm. Typical 100-140 mm. Wider = stiffer in bending and torsion, more wetted area and drag.
Thickness	6 / 16 / 40 mm	Maximum mast thickness (side to side). Limits 6-40 mm, and 4-25 % of Width. Typical 13-19 mm (about 11-15 % of chord). Thinner = less drag and spray; thicker = stiffer and stronger. The notice reports the resulting t/c.
TE thickness	0.1 / 0.8 / 3 mm	Trailing edge thickness of the mast section. Limits 0.1-3 mm, and at most 3 % of Width. Typical 0.6-1.0 mm for wet-laid carbon in printed molds.
Points/surface	10 / 36 / 120	Limits 10-120. Typical 30-40. Cosine-spaced points on each side of the symmetric NACA 00xx mast section.

Taper and top

Label	Min / default / max	Meaning
Top chord	30 / 120 / 400 mm	Chord of the mast at the top, where it meets the plate; Width is the chord at the base. Limits 30-400 mm. Typical equal to Width (no taper), or 10-30 % more for stiffness where the bending moment peaks, or less for a slimmer mast where it pierces the surface.
Taper side	dropdown: Symmetric, Straight leading edge, Straight trailing edge (default Symmetric)	Where the chord change goes. Symmetric: both edges move equally about the mid-chord. Straight leading edge: the leading edge stays vertical and the trailing edge moves. Straight trailing edge: the trailing edge stays vertical and the leading edge moves. Only matters when Top chord differs from Width.
Top thick	4 / 16 / 40 mm	Section thickness at the top of the mast; Thickness is the value at the base. Limits 4-40 mm, and 4-25 % of the local chord. Typical equal to Thickness, or 20-50 % more to carry the bending and torsion where the mast meets the plate. The section stays a NACA 00xx, scaled.
Thicken from %	0 / 70 / 95	Height, as % of Height from the base, where the thickness starts to grow from Thickness toward Top thick, following a smooth S-curve to the top. Limits 0-95 %. Typical 60-80 %: the lower mast keeps its drag-optimised section and only the loaded top end thickens.
Root fillet	0 / 10 / 30 mm	Fillet radius where the mast meets the plate's underside, all round the section. Limits 0-30 mm. Typical 8-15 mm; 0 = sharp corner. Gives the UD caps and the tab plies a radius to turn through into the plate instead of a 90 deg fold, and spreads the peel at the corner where the side-load moment peaks. Held to 2 mm less than the room between the section and the plate edge. The fillet runs round the section's surface and stops at the trailing-edge line (a 0.8 mm edge cannot carry it), leaving a notch of TE thickness there; the console's MAST_FILLET line reports the seam and the attempt used.

Position

Label	Min / default / max	Meaning
Mast X	-300 / -20 / 300 mm	Mast leading edge position along X relative to the wing root pivot (world origin), positive = aft. Limits -300 to +300 mm. Typical -40 to 0 mm, i.e. the mast sits over the forward part of the wing. This is where the rider's weight enters the foil, so compare it with the Foil performance trim CG.
Base height	-150 / 30 / 150 mm	Z of the mast's bottom end, measured from the wing root pivot. Limits -150 to +150 mm; negative is normal when the mast sits in a Foil fuselage pocket, whose floor can be below the pivot. Typical: the value the Foil fuselage warning asks for, so the mast sits on the pocket floor.

Motor pod

Label	Min / default / max	Meaning
Motor pod	checkbox, default off	Adds a streamlined motor pod on the mast (eFoil). Off for pump, SUP, surf, wing, wind and kite foils.
Motor dia	30 / 70 / 150 mm	Pod outside diameter (motor can plus housing). Limits 30-150 mm. Typical 60-90 mm for eFoil motors.
Pod height	0 / 150 / 1200 mm	Height of the pod centreline above the mast base. Limits 0-1200 mm, and the pod must stay within the mast height. Typical 100-250 mm; lower keeps the prop deeper, higher keeps it away from the wing.
Nose start %	0 / 5 / 80	Where the blend starts along the mast chord, as % from the mast leading edge. Limits 0-80 %. Typical 2-10 %: an early start gives the longest, gentlest blend and least drag. The first section matches the mast's own thickness there, so the pod grows out of the mast faces and never protrudes ahead of the mast.
Fairing length	20 / 150 / 400 mm	Length of the blend from the mast section to the full motor diameter. Limits 20-400 mm. Typical 2-3 x Motor dia; longer means a gentler flank angle and lower drag. Must fit between the nose tip and the rear face.
Blend height %	30 / 100 / 100	Height of the first blend section as % of the motor diameter. Limits 30-100 %. Typical 70-100 %. 100 % = the pod starts at full motor height and grows only in width out of the mast faces; lower also tapers it vertically for a softer entry.
Aft extension	0 / 100 / 400 mm	How far the pod extends behind the mast trailing edge, ending in a flat rear face for modelling the motor mount by hand. Limits 0-400 mm. Typical 60-150 mm, enough to clear the mast for the motor and prop.
Cavity dia	5 / 20 / 140 mm	Diameter of the placeholder motor cavity in the pod's rear face. Limits 5-140 mm, and at most Motor dia minus 4 mm (2 mm of pod wall each side). Typical 20 mm as a placeholder, a small locating pocket to open out to the motor by hand. The Foil mold molds the pod's outside with the mast and makes a plug of this cavity, pulled after cure.
Cavity length	2 / 10 / 400 mm	Depth of the placeholder cavity from the pod's flat rear face forward. Limits 2-400 mm; shortened if it would run into the nose blend. Typical 10 mm as a placeholder, then the motor can length plus 5 mm once it is chosen.
Pod tilt	-15 / 0 / 15 deg	Rotation of the whole pod about its centre, nose-down positive. Limits -15 to 15 deg. Typical 5-8 deg for an eFoil: the board rides nose-up at low speed, so a nose-down pod keeps the prop nearer the flow at take-off and its thrust's downward component behind the CG lifts the nose against the thrust-height couple. 0 = axis parallel to the fuselage. Foil performance reports the thrust angle to the flow at every speed.

Plate

Label	Min / default / max	Meaning
Plate length	80 / 165 / 300 mm	Board plate length along X. Limits 80-300 mm. Typical 150-180 mm.
Plate width	60 / 120 / 200 mm	Board plate width across Y; must exceed Track spacing plus a bolt diameter. Limits 60-200 mm. Typical 110-130 mm.
Plate thick	4 / 10 / 30 mm	Board plate thickness. Limits 4-30 mm. Typical 8-12 mm; must be deeper than the countersink.
Track spacing	30 / 90 / 180 mm	Distance between the two rows of bolts (the board's two tracks). Limits 30-180 mm. Typical 90 mm, the common twin-track spacing; check your board.
Bolts/row	1 / 2 / 4	Bolts in each of the two rows. Limits 1-4. Typical 2.
Bolt pitch	10 / 40 / 200 mm	Spacing between bolts along each row, centred on the plate. Limits 10-200 mm. Typical 40-60 mm. Ignored with one bolt per row.
Bolt	dropdown: M4, M5, M6, M8, M10, Custom hole sizes (default M8)	The track bolts. Sets the plate's clearance hole (M6 6.5, M8 8.5 mm: 0.1 mm looser than the stab's close fit, since four holes must line up with the tracks) and the countersink. Typical M8 (most boards), M6 on some. Custom hole sizes: type the hole below.
Bolt dia	3 / 8.5 / 12 mm	Bolt clearance hole diameter (Custom only). Limits 3-12 mm. Typical 8.5 mm for M8, 6.5 mm for M6.
Countersink	checkbox, default on	90 deg countersink for flat-head bolts on the plate's underside (the mast side), to the ISO 10642 head diameter of the chosen Bolt (M6 13.4, M8 17.9 mm; Custom: twice the hole). Bolts go in from below, through the plate, into the board's track nuts, so the heads sit flush with the underside.
Log inputs	checkbox, default on	Prints every parameter on this feature to the FeatureScript console as one line, so the whole set can be pasted elsewhere (e.g. back to an AI assistant).

Foil fuselage

Body

Label	Min / default / max	Meaning
Parallel aft	0 / 40 / 300 mm	How far the full-diameter body continues behind the mast trailing edge before it tapers. Limits 0-300 mm. Typical 20-60 mm.
Tail dia	6 / 24 / 60 mm	Body diameter under the stab, after the taper. Limits 6-60 mm; clamped to the body diameter. Typical: equal to Tube dia (no taper before the stab - the fuselage is a high-stress part and the stab block needs a full body under it), tapering only in the tip. Smaller only for a lightly loaded foil.
Overhang	-200 / 60 / 300 mm	Where the fuselage ends: the X of its pointed tip, measured from the stab root trailing edge. Positive = behind the stab, negative = the tip ends under the stab. Limits -200 to +300 mm. Typical 40-80 mm (tip fully behind the stab, untouched by the seat).
Tip length	5 / 40 / 200 mm	Length of the elliptical tip, from Tail dia down to a point. Independent of Overhang, so the tip is always a full ellipse. Limits 5-200 mm. Typical 2-3 x Tail dia; longer is lower drag.

Stab seat

Label	Min / default / max	Meaning
Stab block	checkbox, default on	A rectangular block as wide as the stab's Mount pad and as long as the seat, between the body axis and the pad plane, so the seat is the full pad width whatever the tube diameter and the tapped holes have solid material. With Pad side Top it hangs under the body (the fuselage passes over the stab); with Bottom it stands on top. Off = the seat is just a flat cut on the round tube.
Seat margin	0 / 3 / 30 mm	Extra seat (and block) length beyond the stab root chord at each end (never past the fuselage tip). Limits 0-30 mm. Typical 2-5 mm.
Stab bolt dia	3 / 6.8 / 10 mm	Tapping drill for the stab bolts, used ONLY when the Foil stabiliser's Bolt is set to Custom hole sizes; with a named bolt (M4-M10) the drill comes from the stab (M6 5.0, M8 6.8 mm) and this field is ignored. Limits 3-10 mm. Typical 6.8 (M8) or 5 (M6). The positions come from the stab's Mount bolts (Bolt 1 at, Bolt pitch), so the two always match.
Bolt depth	3 / 14 / 40 mm	Depth of each blind tapped hole from the seat plane into the body. Limits 3-40 mm. Typical 1.5-2 x bolt diameter. With the fuselage over the stab (Pad side Top) the hole goes up: the depth available is the pad plane's drop below the axis plus the local radius, less 1 mm of wall (about 15 mm on a 24 mm tail seated 4 mm under the axis). A hole that would break through is skipped with the fix named.

Mast pocket

Label	Min / default / max	Meaning
Mast pocket	checkbox, default on	Cuts a pocket in the top of the fuselage for the mast's bottom end, shaped to the Foil mast's own section plus Clearance. Its position AND depth follow the mast: the floor is at the mast's Base height, so moving the mast reshapes the pocket. Needs a Foil mast feature above this one.
Clearance	0 / 0.2 / 2 mm	Gap between the mast and the pocket walls, all round. Limits 0-2 mm. Typical 0.1-0.3 mm for a bonded or bolted fit.
Wing pin holes	checkbox, default on	Holes through the spigot matching the wing socket's Pins, Pin bolt and Pin pitch, at the same clearance size.
Log inputs	checkbox, default on	Prints every parameter on this feature to the FeatureScript console as one line, so the whole set can be pasted elsewhere (e.g. back to an AI assistant).

9. Console log reference

Turn on *Log inputs* on a feature to print its inputs to the FeatureScript console. The keys are stable so a log can be pasted back to an assistant or into the scripts.

Key	Feature	Content
WING_INPUTS , STAB_INPUTS , MAST_INPUTS , FUSELAGE_INPUTS , PERFORMANCE_INPUTS , MOLD_INPUTS	each	every input of the feature, with its version as the first field
FUSELAGE_SEAT	fuselage	the seat and the tapped holes made, or the conflict and the fix

Key	Feature	Content
MAST_FILLET	mast	the seam edges found at the plate and the fillet attempt that worked
===== FOIL PERFORMANCE ===== block	performance	the full report, the propulsion block and the CSV
UD_CAP_PLIES	mold	per-cap cut list: ply, length, width at root, width at end
UD_TABS	mold (mast)	tab plies per face, size, fold position, interleave
LAYUP	mold	volumes, masses, plies, tow, resin, what to buy
STIFFNESS	mold	deflection, EI, cap stress; mast: stress at the plate and at its worst point, bolt tension, junction, plate
STAB_CONES	mold (stab)	the bolt cones made and their positions
MAST_PLUG	mold (mast)	the pod plug's cores and depth
TEMPLATES	mold	the order of shapes on the template plate and their cut counts

Each feature's version is printed as the first field of its log line; the file's build number is in the Feature Studio header. If a pasted log shows a version older than the file you have, the Feature Studio was not pasted over.

10. Model limits, versioning and the AI assistant

10.1 Model limits

- Results are best for comparing variants. Profile drag is about $\pm 20\text{--}30\%$ absolute; induced drag within $5\text{--}10\%$ for modest sweep.
- With the analytic section model, stall comes entirely from the Section CL max input; use a polar for real behaviour.
- **Stall speed is optimistic.** It is the speed at which the most loaded section of a perfect, clean section reaches its 2D CL max in dead steady flight. Hand finish, the fuselage junction, wake, free surface and the margin a rider needs all lower the usable CL max. Stall speed goes with $1/\sqrt{\text{CL max}}$, so calibrate: take one foil you know, find the Section CL max (or polar CL max) that reproduces its real take-off speed, and use that value for new designs.
- **One section family per surface.** Thickness ratio varies root to tip and Min tip holds an absolute floor, but camber and profile are the same along the span. Blending a cambered root to a flatter tip is not modelled yet.
- Lifting line ignores sweep and anhedral except for moving the aerodynamic centre; strongly swept or drooped wings will differ.
- Downwash at the stab is the classic far-field estimate.
- Not modelled: free-surface effects beyond the two MAC warning, ventilation, cavitation onset (only the cavitation number is reported), pumping and unsteady motion, board and rider aerodynamics, ploughing before lift-off, acceleration, junction drag between mast and fuselage, pod base drag.

- Propulsion is the B-series open-water estimate, about $\pm 20\%$, with fixed motor losses; no wake, no prop cavitation, no thermal limits.
- Structure: a spar-cap estimate with a simple EI model; no shear, torsion, buckling, fatigue, core or insert design; the junction check ignores peel at the fillet and the tabs' own anchoring.

10.2 Versioning

The Feature Studio header carries the build number and every feature's version. The build number goes up on every delivery; a feature's version goes up only when that feature changes, by 0.001 for an ordinary change and to the next whole number for a redesign. A changelog comment above the version constants records every change and why. The `toolVersion` field in each console log line tells you which version produced it.

10.3 The AI assistant and its scripts

The foil design assistant is a skill for AI assistants that reads what the features print and helps you decide what to change next. Paste your console log and it explains the results, checks them against the targets in chapter 5, and recommends a few changes in the dialog's own labels, with the expected effect and the cost. Then it compares the next run with the last.

Assistant	Setup
Claude (desktop app, claude.ai, Claude Code)	Add <code>foil-design-assistant.skill</code> . It loads when you paste foil output and runs the scripts itself.
ChatGPT	Use <code>foil-design-assistant-portable.md</code> as a custom GPT's instructions, with <code>parameters.md</code> and the scripts as knowledge, and Code Interpreter on. Or attach both files to a chat.
Other assistants	Give them <code>foil-design-assistant-portable.md</code> and <code>parameters.md</code> at the start of the chat. Without code execution they follow the manual method in the file.

What to paste. Turn on *Log inputs* on every feature, regenerate, and copy the whole FeatureScript console, plus the notice of any feature showing a warning. One paste then carries every input and output, so every recommendation can be checked.

The scripts. Four Python 3 scripts come with the assistant, standard library only. An assistant that can run code runs them for you. To run them yourself, save the console to a text file and use a terminal:

```
python3 analyse_sweep.py run.txt [previous_run.txt] python3 pitch_sweep.py run.txt --vmin 15
python3 polar_summary.py polar_re1.txt polar_re2.txt python3 cut_templates.py run.txt
templates.dxf
```

- `analyse_sweep.py` summarises a run: best L/D, minimum power, take-off loading, stance shift, static margin, drag split and, with propulsion, the lowest Wh/km, throttle range, top speed and lowest powered margin. Given two runs, it compares them.
- `pitch_sweep.py` rebuilds the design's trim from the log, checks it against the tool's own numbers, and finds the smallest stab, arm and decalage that reach a target static margin, including the effect of moving a motor pod.
- `polar_summary.py` turns one or two Xfoil or XFLR5 polar files into the thirteen numbers the Section polar group takes.
- `cut_templates.py` writes the skin and UD ply outlines as a DXF from the log, for cutting without Onshape.